How Costly Is Carbon Abatement in the Residential Sector? Evidence from Energy Efficiency Obligations

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Abstract

We evaluate the French Energy Savings Certificates program, under which energy retailers subsidize building retrofits to meet mandatory efficiency targets. Using sub-municipal electricity and gas data from 5 million housing units (2018–2021), we first estimate its impact on energy use and associated carbon emissions. We then develop a novel revealed-preference approach to measure the average carbon abatement cost, leveraging the tradability of certificates. Combining certificate prices with estimated energy savings and emissions factors yields a preferred estimate of €181 per ton of CO_2e. Accounting for the pass-through of program costs into retail energy prices and the resulting reduction in energy demand lowers this to €54 per ton, indicating overall cost-effectiveness. This is despite official reports overstating realized energy savings by 69%.

Keywords: Energy efficiency, energy efficiency obligations, abatement cost of carbon

JEL codes: L78, Q48, Q58

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1 Introduction

Improving the energy efficiency of the building stock is a cornerstone of the energy transition. According to the International Energy Agency, global investment in building energy retrofits averaged more than USD 160 billion annually between 2015 and 2021 (IEA 2022). This represents around 60% of all global energy efficiency investments, more than twice the amount devoted to the transportation sector. And much more is expected in the years ahead. In the European Union, the Renovation Wave strategy aims to double the annual renovation rate in order to support the achievement of the 55% greenhouse gas emissions reduction target by 2030 compared with 1990 levels. Its implementation requires allocating a lion's share to energy efficiency investments in buildings. In France, they are expected to account for around 75% of the total investments necessary to meet the national climate target (Pisany-Ferry and Mahfouz 2023).

One of the central questions of climate policy is where to reduce emissions across the economy. In this respect, the prominent role assigned to building renovation in decarbonization strategies stems from the recognition of its multiple co-benefits, including job creation, health gains, reduced energy poverty, and greater energy independence. It also reflects strong confidence among policymakers and stakeholders in its energy impacts. Since the late 1970s, insulation measures and the adoption of energy-efficient heating and cooling technologies have been widely promoted as a win-win strategy to reduce energy consumption and associated carbon emissions. This perspective has largely been grounded in engineering model projections suggesting that energy savings would be sufficiently large, over time, to more than offset upfront investment costs. The most prominent illustration is the so-called McKinsey marginal abatement cost curve, which claims that energy efficiency investments in buildings exhibit negative carbon abatement costs (McKinsey 2009).

This view has been challenged by a growing empirical literature documenting that realized energy savings systematically fall short of engineering projections (Gerarden, Newell, and Stavins 2017). In the United States, recent studies estimate the energy performance gap between projected and realized savings to average around 55% (see e.g. Allcott and Greenstone 2024; Fowlie, Greenstone, and Wolfram 2018; Christensen et al. 2023; Zivin and Novan 2016). Papineau, Rivers, and Yassin (2025) report similar magnitudes for Canada.[1]

1. Related studies examine similar issues for other technologies and energy-using durables, including smart metering (Brandon et al. 2022), household appliances (Boomhower and Davis 2014; Davis, Fuchs, and Gertler 2014), and the energy performance of newly built housing (Davis, Martinez, and Taboada 2020; Levinson 2016; Kotchen 2017).

Lower than expected realized energy savings directly translate into higher carbon abatement costs than previously thought. Using their energy impact estimates, Fowlie, Greenstone, and Wolfram (2018) derive a cost of USD 350 per tCO_2e for the Weatherization Assistance Program (WAP) in Michigan[2]. Similarly, Lang and Lanz (2022) estimate an abatement cost of USD 380 per tCO_2e for a Swiss energy efficiency program. In contrast, Christensen et al. (2023) find a substantially lower cost, on the order of USD 7.6 per tCO_2e.

These abatement cost figures are obtained by calculating the private cost of reducing emissions as the difference between upfront investment expenditures and the discounted stream of realized energy bill savings over the lifetime of the measures. Avoided carbon emissions are then obtained by multiplying realized energy savings by the relevant fuel-specific emissions factor. Unsurprisingly, as the authors acknowledge, the resulting estimates are highly sensitive to assumptions regarding investment lifetimes and discount rates. For instance, Christensen et al. (2023) show that reducing the assumed lifetime of measures from 30 to 20 years increases estimated abatement costs by a factor of six. Similarly, Fowlie, Greenstone, and Wolfram (2018) document a 40% increase when the discount rate is raised from 3% to 7%.

In this paper, we estimate the carbon abatement cost implied by the French Energy Savings Certificates (CEE) program, using a novel revealed-preference approach that departs fundamentally from this accounting-like approach. The program imposes energy-savings obligations on energy suppliers, who mainly comply by subsidizing household energy-efficiency investments. Rather than summing assumed investment costs and discounted energy savings, we exploit the tradability of certificates: their market price, combined with per-certificate carbon savings, reveals the cost of abatement.

The analysis proceeds in four steps. First, we estimate the causal effect of the program on residential gas and electricity consumption using sub-municipal data. The treatment variable is defined at the neighborhood–year level as the number of energy savings certificates issued for energy efficiency investments completed in that year. Each neighborhood corresponds to an IRIS as defined by the French National Institute of Statistics and Economic Studies (INSEE)[3].

2. This value is taken from Table 1 in Gillingham and Stock (2018), where the authors survey empirical estimates from the literature of the private cost per avoided ton of CO_2e. When accounting for benefits from reduced local environmental externalities, as done by Fowlie, Greenstone, and Wolfram (2018), this cost decreases to USD 200 per tCO_2e .

3. IRIS stands for *Îlot Regroupé pour l'Information Statistique* in French, i.e., aggregated unit for statistical information. It is the fundamental unit for dissemination of infra-municipal data in France, which gathers around 2,000 residents.

The data covers 546,674 investments across 4,671 neighborhoods over the period 2018–2021. Because certificates are issued in proportion to projected lifetime energy savings, their estimated impact on energy use directly identifies the gap between these projections and ex-post realized savings.

Identification relies on an instrumental variables strategy exploiting the effect of past temperature shocks on the salience of heating- and cooling-related housing characteristics, which in turn affects households' investment decisions. Our instrument is the one-year lagged annual heating degree days (HDD). The length of the lag is assumed to roughly correspond to the average delay between investment decisions and retrofit completion. The identifying assumption is that lagged temperature shocks are orthogonal to contemporaneous changes in energy consumption. It is supported by the inclusion of municipality and year fixed effects, which absorb time-invariant climatic differences and common shocks, mitigating concerns about serial correlation in temperature-driven energy demand.

Our preferred estimates indicate that investments supported by the CEE program reduce annual neighborhood-level gas and electricity consumption by 1.36% on average. This realized reduction represents 31% of the energy savings predicted by the engineering estimates used to issue certificates. Furthermore, combining the estimated energy savings with fuel-specific carbon emission factors, we find that the program reduces annual carbon emissions by 1.64%.

In a second step, we develop a formal model of the program to carefully examining how certificate prices translate into the private costs borne by both obligated parties and subsidized households. The model describes a two tiered market structure where energy retailers seeking to minimize their compliance costs compete to source eligible investments and offer financial incentives to households on the primary tier, while they exchange certificates in a secondary market. Under a set of linearity assumptions and assuming competitive markets, we derive a formula that yields the average cost of carbon from the certificate price.

In a third step, the formula is calibrated using the energy savings estimated in the first stage, the corresponding carbon emission factors, and the certificate prices observed during the study period. Our preferred estimate is a cost of €181/tCO_2e.[4]

In a fourth step, we examine an additional channel through which the program reduces energy use. The grants offered to households, together with the associated commercial and administrative

4. All monetary values referring to the French program are expressed in 2019 euros hereafter

costs, are ultimately financed through marginal increases in retail energy prices (Rosenow, Cowart, and Thomas 2019). These price increases create incentives for all energy users—not only those investing in energy efficiency—to reduce energy consumption and carbon emissions. Combining our results with outside estimates, we show that this energy price effect dominates the investment subsidy channel, accounting for 73.1% of the total energy savings generated by the program. Because the implicit carbon price associated with the energy price increase is only €8/tCO_2e, the program's average abatement cost consequently falls substantially to €54/tCO_2e.

The paper proceeds as follows. In section 2, we discuss how the paper relates to the existing literature. Background information on the French program is presented in 3. We introduce the data used in the analysis in section 4. The empirical strategy is exposed in section 5. Section 6 provides regression results and some robustness checks. We turn to the estimation of the abatement cost of carbon in the following section, and we conclude in Section 8.

2 Relation to the literature

This paper contributes to the literature in three main ways. Our first contribution is the evaluation of an energy efficiency program with a distinct and institutionally specific design. The program is based on energy efficiency obligations imposed on multiple energy retailers that compete to source the least-cost investments. This design stands in contrast to US-based programs. So-called Energy Efficiency Resource Standards (EERSs) generally impose obligations on local distribution monopolies regulated by state public utility commissions, with compliance costs recovered through regulated charges paid by all customers connected to the network (for evaluations, see Aroonruengsawat, Auffhammer, and Sanstad 2012; Arimura et al. 2012). By contrast, European programs rely on obligations imposed on retailers operating in liberalized and competitive energy markets (Giraudet, Glachant, and Nicolaï 2020). Although these programs constitute a cornerstone of the EU energy policy, evaluation studies are scarce: Vivier and Giraudet (2024) evaluate the French program with a calibrated bottom-up model; Rosenow and Bayer (2017) conduct an EU-wide cost–benefit analysis based on policy reports and previous studies. To the best of our knowledge, we provide the first empirical evaluation of such programs.

A further distinction is that the evaluation explicitly accounts for the pass-through of program costs into retail energy prices and the resulting effects on energy demand. Incorporating this channel substantially alters the assessment of the program's cost-effectiveness, reducing the estimated carbon abatement cost by a factor of three.

Second, we provide a new estimate of the energy performance gap between projected and realized savings. Our estimate of a realization rate of 31% falls within the range of previous evaluations (e.g., Fowlie, Greenstone, and Wolfram 2018; Christensen et al. 2023; Lang and Lanz 2022; Papineau, Rivers, and Yassin 2025). From a methodological perspective, one originality is the use of a continuous treatment variable, the number of energy efficiency certificates. As result, the associated energy impact is more homogeneous than a binary treatment invest/no invest used in previous studies. This property is useful as it potentially mitigates the biases generated by standard quasi-experimental estimations (see the arguments by Chaisemartin and d'Haultfœuille 2020).

Third, we propose a novel revealed-preference approach to estimating the cost of CO_2 abatement that exploits the tradability of program obligations. Relative to the standard accounting-like approach, this method captures a broader set of costs and benefits associated with the program.

On the obligated-party side, the certificate price reflects the full cost of certificate production, including the subsidy granted to households as well as the costs of persuading households to participate, which can represent a substantial share of total program costs. For instance, Fowlie, Greenstone, and Wolfram (2018) report that, in a Michigan field experiment, expenditures on outreach, information provision, and promotional activities accounted for approximately 20% of retrofit costs. The certificate price also incorporates the administrative costs associated with processing energy-efficiency investments into certificates.

On the household side, competition among obligated parties to trigger the least-cost investment opportunities leads to endogenously determined grant sizes, allowing us to infer households' willingness to accept (WTA) investing in energy efficiency. The WTA covers all private costs and benefits as perceived by households when making their decision. These include non-monetary costs incurred at the investment stage, such as the time and cognitive effort required to gather information, coordinate with contractors, and manage disruptions during the retrofit process. Evidence by Fowlie, Greenstone, and Wolfram (2015) on encouragement costs indirectly suggests that such "hassle costs" are substantial. Houde et al. (2025) provide direct experimental estimates in the context of Canada's energy coach program, valuing households' time spent at the investment stage at approximately USD 370 for a heat pump installation. Our framework also accounts for post-retrofit non-monetary gains, notably improvements in comfort arising from increased use of energy services through rebound effects, which are typically ignored in the literature when calculating abatement costs A notable exception is Fowlie, Greenstone, and Wolfram (2018), who directly measure changes in indoor temperatures to capture such benefits. While we do not separately identify these non-monetary costs and benefits, our approach allows them to be incorporated into the abatement cost calculation rather than implicitly assumed away.

In addition, inframarginal investments are excluded from the calculation. Accordingly, we do not account for the abatement cost impact of all investments supported by obligated parties, but only for those investments that would not have been undertaken in the absence of this financial support. This approach improves the policy relevance of the findings.

More generally, the analysis also relates to the behavioral public economics literature. Following the terminology of Chetty (2015), our revealed-preference approach captures ”decision utility”, both for obligated firms choosing the intensity of their commercial efforts and the level of energy-efficiency grants, and for households deciding whether to invest. By contrast, the accounting-style methodology discussed in the introduction aims to approximate ”experience utility”.

In the energy efficiency literature, measuring the distance between the two and its sources is the central challenge of the well-developed literature on the ”energy efficiency gap” (for a survey, see Gerarden, Newell, and Stavins 2017). This measurement has crucial policy implications: experience utility is considered to reflect households' actual well-being and serves as the benchmark for implementing paternalistic public interventions (Allcott 2011; Allcott, Mullainathan, and Taubinsky 2014).

However, as argued by Bernheim and Taubinsky (2018), this inevitably requires theoretical assumptions to externally define the private optimum. In the specific context of estimating carbon abatement costs from home energy retrofits, current practices typically rely on a series of strong assumptions, including zero non-monetary costs and benefits or expectations about future energy prices—often proxied by current prices. Results are also highly sensitive to assumptions regarding investment lifetimes and discount rates. For instance, Christensen et al. (2023) show that reducing the assumed lifetime of measures from 30 to 20 years increases estimated abatement costs by a factor of six. Similarly, Fowlie, Greenstone, and Wolfram (2018) document a 40% increase when the discount rate is raised from 3% to 7%.

There is no simple solution to the problem. Measure lifetimes are not directly observed in the data and therefore rely on engineering assumptions, while the choice of discount rate remains theoretically and empirically challenging. For example, Houde and Myers (2019) assume that privately rational discount rates lie between the annual percentage rate on credit cards (12%) and the three-year return on Treasury bonds (2%). This interval is arguably wide.

Further progress requires disentangling measurement error by the observer from cognitive biases that prevent households from making privately optimal decisions. In this respect, La Nauze and Myers (2023), who study whether consumers optimally acquire information about energy costs in appliance markets, point to promising directions for future research on energy retrofits.[5].

As with any empirical evaluation, our approach relies on exogenous assumptions, especially concerning the functioning of the certificate market. Nevertheless, it provides a useful alternative benchmark for assessing the cost-effectiveness of retrofit programs, one that complements existing approaches based on experienced utility and highlights the robustness of policy conclusions to different evaluative frameworks.

3 Institutional background

Energy efficiency obligations have become a widely used policy instrument to promote end-use energy savings across a broad range of jurisdictions. As of 2022, the International Energy Agency identified at least 48 active programs operating in 31 countries worldwide (IEA 2022). Within the European Union, the Energy Efficiency Directive 2023/1791 establishes energy efficiency obligations as a core mechanism for achieving national energy consumption reduction targets. In the United States, 24 states operate so-called Energy Efficiency Resource Standards (EERSs)[6]. They are energy efficiency obligations primarily imposed on local distribution monopolies regulated by state public utility commissions, with compliance costs recovered through regulated charges paid by all customers connected to the network. In contrast, European programs are obligations imposed on a set of energy retailers operating in liberalized and competitive markets. Comparable schemes have also been implemented in countries such as Australia, Brazil, Canada, China, South Korea, South Africa, and Uruguay (Crampes and Léautier 2020).

The French CEE program[7] is the largest in Europe with a total investment of nearly €4 billion

5. This issue has already been studied in other domains, including tax filing (Benzarti 2020), individual health insurance (Drake, Ryan, and Dowd 2022), and drug prescription (Currie and Musen 2025)

6. This count includes only programs with mandatory energy efficiency requirements. Two additional states have voluntary targets, and two states allow energy efficiency as a compliance option under their renewable portfolio standards. See https://www.epa.gov/statelocalenergy/energy-and-environment-guide-action.

7. CEE stands for "Certificats d'Economie d'Energie", Energy Savings Certificates in english.

per year (Broc, Stańczyk, and Reidlinger 2020). It supports retrofit works in the residential, industrial, or tertiary sector, and to a lower extent in agriculture and transport. The residential sector gathers more than two thirds of the total energy savings (DGEC 2022).

The mechanism works as follows. Every four years, individual energy savings targets are assigned to retailers of electricity, gas, and gasoline in proportion to their sales. In practice, obligated suppliers predominantly meet their targets through the provision of grants for residential energy retrofitting. For each investment subsidized, they receive a certain amount of energy savings certificates from the regulator upon providing proof of investment (e.g., an invoice for the installation of insulation). The number of certificates awarded is designed to reflect the discounted stream of expected energy savings over the lifetime of the measure. At the end of each compliance period, suppliers must return enough certificates to cover their obligation.

The standardized calculation of certificates is based on ex ante engineering estimates. In the residential sector, efficiency measures are grouped into 28 categories, such as roof insulation and high-efficiency individual boilers (see Table A2). Each category has a formula linking savings to project size and location. For example, installing 50 m^2 of roof insulation in the coldest French climate zone saves 85,000 kWh over 30 years, while a high-efficiency boiler in the same zone saves 24,800 kWh over 17 years. Lifetime savings are discounted at a 4% annual rate, and each certificate represents 1 kWh of expected savings.

The presence of multiple obligated firms implies competition in sourcing energy efficiency investments that minimize the cost of producing compliance certificates. The program grant substantial flexibility regarding both the type and location of eligible investments. Obligated energy retailers may target their own customers or other energy users, including customers supplied by competing retailers (Giraudet, Glachant, and Nicolaï 2020). They also freely choose the level of grants offered to induce these investments. Compliance costs are ultimately recovered through market adjustments in retail energy prices. In this way, the program do not increase public expenditures as utilities pass on their costs to consumers (Rosenow, Cowart, and Thomas 2019). While the resulting prices increase contributes to reduced energy consumption, this indirect effect is not counted toward compliance with the program. We quantify its magnitude in subsection 7.3.

An additional source of flexibility arises from the possibility for obligated firms to meet part or all of their obligation by purchasing energy savings certificates from other obligated parties. Trading volumes reached 179 TWhc in 2018 and 288 TWhc in 2019, highlighting the central role of certificate trading in meeting the annual obligation of 533 TWhc. Certificate prices reflect both the underlying marginal cost of generating energy savings and the stringency of the obligation, a feature we exploit to infer the carbon abatement cost implied by the program.

The emphasis on cost minimization is further reflected in the relatively limited requirements for project implementation. Unlike other energy efficiency programs such as the Weatherization Assistance Program in the United States or Germany's Federal Subsidy for Efficient Buildings, the French program does not mandate pre-retrofit energy audits or ex post certified quality controls. To limit inframarginal renovations, obligated suppliers are only required to submit to the regulator an affidavit in which the beneficiary household declares that the financial support was decisive for its investment decision.

Last, the program has also been helping to combat fuel poverty thanks to two social equity provisions: a sub-obligation whereby energy retailers are required to devote at least 25% of their energy-saving operations to low-income households, and a bonus system that doubles the amount of certificates issued for the energy renovation of homes occupied by households in the bottom quartile. The bonus system has mechanically created a gap between realized energy savings and the number of certificates. Obviously, these bonus certificates are not included when the government reports the quantity of energy savings achieved through the program, and we likewise exclude them from the dataset used in the analysis. We will take the existence of these public subsidies into account in our abatement cost calculations.

4 Data

4.1 Data sources

The analysis builds on three primary data sources: residential electricity and gas consumption data, certificate data detailing the quantity and characteristics of energy retrofits subsidized by the program, and weather and city-level socio-demographic data.

Residential electricity and gas use. The data is provided by the *Opérateurs des Réseaux d'Energie* agency. It gathers electricity and natural gas residential consumption in France at the infra-municipal level over 2018-2021. Each observation corresponds to the annual consumption of an aggregated unit for statistical information (IRIS) as defined by the French National Institute of Statistics and Economic Studies – INSEE Institut national de la statistique et des études économiques (INSEE) 2016. It is the fundamental unit for dissemination of infra-municipal data in France, which gathers around 2,000 residents[8].

Energy retrofits. The Energy Efficiency Obligation database is hosted by the Centre d'Accès Sécurisé aux Données (CASD), a French public-interest organization that provides secure access to confidential data for non-profit research purposes (CASD 2023). The dataset includes more than 6 million energy efficiency operations carried out in the French residential sector between 2015 and 2024. We focus on the years 2018-2021, which correspond to the program's fourth obligation phase[9]. The dataset provides detailed information on the types of work undertaken, the dates of initiation and completion, the expected lifespan of each retrofit, its geographic location (at the city level), and the number of certificates issued.

This information allows us to estimate the projected energy savings associated with each subsidized retrofit. The calculation begins by converting the number of certificates into lifetime energy savings, after excluding any bonus certificates. These lifetime savings are then transformed into annual savings using data on the expected lifespan of each measure and by applying the standardized 4% discount rate. Since the completion month is known for every renovation, savings are accounted for only from the month following completion. When aggregated at the municipal level, this procedure ensures that annual savings accurately reflect the actual timing of renovation activities.

A key limitation is that these data cannot be matched to household-level energy consumption. Consequently, our empirical analysis is conducted at the infra-municipal level, implying that both treatment and outcomes are measured in aggregate rather than at the individual household level.

Weather data and sociodemographics. Heating and cooling degree are standard engineering metrics used to approximate heating demand, as they capture the influence of temperature and humidity on energy consumption. The data are provided by Météo France, the French national meteorological service, on a grid of 9,892 cells, each covering 64 km^2 (8 km $\times$ 8 km). For each grid cell and day, the average temperature is recorded. HDD and CDD values are then computed

8. Towns with more than 10,000 inhabitants, and a large proportion of towns with between 5,000 and 10,000 inhabitants, are divided into several IRIS units.

9. This phase was unexpectedly extended by one year due to the COVID-19 pandemic.

at the municipal level, using 17°C (respectively, 23°C) as the reference temperature. The detailed matching procedure is described in Appendix A.

4.2 Study sample.

The data do not provide information on energy consumption from other sources than gas and electricity (e.g., heating oil, liquefied petroleum gas, or district heating), and the retrofit data do not indicate the type of fuel used in retrofitted dwellings. To address this issue, we rely on the French population census (INSEE 2018) and restrict our analysis to IRIS where at least 90% of residents rely on either electricity or natural gas as their primary heating source. In practice, this selection removes mostly rural neighborhoods, which rely relatively more on heating oil, propane, or biomass[10]. The final estimation sample consists of 4,671 neighborhoods, representing 11.66 million residents. Table 1 presents summary statistics for the resulting sample. Compared to the French average, it gathers neighborhoods with a +31% higher per capita natural gas annual consumption, twice more housing units and a 70% higher population.

Table 1: Descriptive statistics

	Estimation sample (N = 4,671)		France (N = 44,093)	
	Mean	SD	Mean	SD
Panel A: Energy use				
Annual electricity use per capita (kWh)	2,168.97	1,039.03	2,860.39	1,101.07
Annual gas use per capita (kWh)	3,582.30	3,306.43	2,725.13	2,426.73
Panel B: Retrofit works				
Projected lifelong energy savings per capita (kWh)	1,981.14	4,680.51	3,281.45	3,795.30
Annual projected energy savings per capita (kWh)	123.68	313.44	197.47	237.20
Panel C: Demographics				
Number of housing units	1,248.87	554.07	584.41	586.51
Population size	2,490.62	1,036.43	1,451.60	1,277.55
Panel D: Weather				
HDD	1,871.72	383.17	2,090.10	450.71
CDD	61.88	56.48	43.80	42.71

10. See Table A1 for a detailed breakdown by degree of urbanization.

Table 2 displays the shares of the main retrofit categories observed in our sample. Investment in building envelope thermal insulation accounts for more than 80% of projected savings and most of these savings are achieved in homes heated by natural gas.

Table 2: Share of projected savings, by type of energy retrofit

Fiche	% projected savings	% in gas-heated dwellings
Roof insulation	38.75	92.46
Wall insulation	20.92	90.79
Floor insulation	18.97	96.48
High EE individual boiler	13.34	n.a.
Window w/ insulated glazing	4.03	88.00

Notes: Table A2 in the Appendix gives the full distribution of the 28 work categories.

5 Empirical strategy

5.1 Model

To estimate the impact of retrofit works supported by the program on residential gas and electricity, let us begin with a simple Two-Way Fixed-Effects (TWFE) model which leverages year-to-year within-neighborhood variations in energy use and energy efficiency investment[11]. The baseline estimating equation writes as:

$$Y_{i,t} = \beta\, X_{i,t} + \lambda \mathbf{W}_{i,t} + \mu \mathbf{D}_{i,t} + \alpha_i + \gamma_t + u_{i,t} \tag{1}$$

The outcome variable $Y_{i,t}$ represents the sum of residential electricity and gas consumption in neighborhood i during year t. Combining these two energy sources allows us to account for potential substitution effects (Giraudet, Houde, and Maher 2018), such as the replacement of a gas boiler by a heat pump. Neighborhood fixed effects are denoted as α_i. We also include year fixed effects γ_t which controls for common shocks influencing energy consumption and the supply of energy savings investments.

11. Each neighborhood corresponds to an aggregated unit for statistical information (IRIS) as defined by the French National Institute of Statistics and Economic Studies (INSEE). See 4.

The variable $X_{i,t}$ represents the cumulative projected savings achieved through works completed until year t. The underlying idea is that the energy savings in a given year are determined by the stock of previous investments, reflecting the persistence of treatment status once a dwelling is retrofitted. In this specification, the coefficient β directly indicates the share of projected savings that are realized ex-post. Note that our continuous treatment variable $X_{i,t}$ differs from the binary treatment variable employed in household-level studies, which captures the occurrence of an energy efficiency investment. A key advantage of this specification is that it mitigates treatment effect heterogeneity across neighborhood–year cells (i, t), which is a known source of bias in two–way fixed–effects designs as discussed by Chaisemartin and d'Haultfœuille (2020). At the same time, it helps reduce the gap between the local average treatment effect (LATE) identified in our empirical strategy and the average treatment effect. We return to these points in the results section.

Due to data limitations for energy use, the variable X includes only projected savings from investments completed from 2018 onwards. Earlier investments are however implicitly captured through the inclusion of neighborhood fixed effects.

Another issue is that the impact of an investment made in year t on the quantity of energy consumed over the entire year depends on the timing of its completion. An investment finalized earlier in the year contribute more substantially to Y_{it} as they are active over a greater portion of the year. To account for this heterogeneity in exposure, projected savings from investments completed during year t are weighted by the proportion of heating degree days occurring after the completion month, relative to the total HDD observed over the entire year. For instance, one unit of projected savings from an investment completed at the end of June is weighted by the ratio of HDD between July and December to the total HDD for the entire year.

Last, the equation includes a vector $\mathbf{W}_{i,t}$ of weather variables (contemporaneous annual HDD and CDD) that influence energy use and may indirectly affect the pace of energy efficiency investments, and a vector $\mathbf{D}_{i,t}$ of demographic controls including the number of housing units heated through electricity or natural gas.

5.2 Instrumental Variation

Self-selection is the main threat to identification. Households choose whether to renovate their homes, and unobserved time-varying factors—such as anticipated demand shocks—that influence this decision can bias results if they are correlated with energy consumption. A family expansion, upcoming retirement, or non-energy-related home improvement may simultaneously increase the likelihood of investing in retrofits and heating needs. In the UK, Peñasco and Anadón 2023 find that glazed extensions added to homes at the time of a retrofit cancel out the benefits of cavity wall insulation. Failing to account for such shocks can lead to an underestimation of energy savings.

Aggregating individual behavior to the infra-municipal level attenuates omitted-variable bias only insofar as unobserved individual shocks are idiosyncratic and therefore average out within neighborhoods. This assumption is unlikely to hold. In our illustration, rises in heating needs are unlikely to be randomly distributed across residents of the same IRIS—there might be a common component (e.g., neighborhood's housing stock sharing the same vintage and deteriorating simultaneously, or correlated household life-cycles such as a locally ageing population) so its effects need not cancel in aggregate.

A further limitation is that retrofits undertaken without support from the CEE program are not observed, as they do not generate certificates. This issue also applies to most previous studies (Allcott and Greenstone 2024; Christensen et al. 2023; Fowlie, Greenstone, and Wolfram 2018; Myers 2020; Papineau, Rivers, and Yassin 2025) which only account for renovations supported by specific energy efficiency programs. While this omission may appear unimportant given that our focus is on the program's impact rather than that of energy efficiency investments, CEE-supported and non-CEE retrofits are likely interrelated, being influenced by similar unobserved factors. This interdependence may potentially introduce bias.

Our solution is to instrument the year-to-year change in projected savings $X_{i,t}$. We exploit infra-municipal data on degree days one year before the investment in energy efficiency is completed. More specifically, the instrument is

$$Z_{i,t} = HDD_{\,i,t-1}$$

The assumption is that annual deviations from the average HDD increase the salience of heating- and cooling-related home characteristics, thereby influencing the decision to invest in energy efficiency. The one-year lag is considered a reasonable estimate between the time the investment decision is made and the completion of the retrofits. Exogeneity arises from the fact that past temperature and precipitation shocks are not correlated with changes[12] in energy use one year later

12. The focus on *changes* in energy use is crucial for identification, as previous temperatures and precipitation may have a persistent effect on energy use behavior.

through channels other than energy efficiency investments. The inclusion of municipality and year fixed effects allows time-invariant climatic differences and common shocks to be absorbed, limiting concerns related to serial correlation in temperature-driven energy demand.

The presence of non-CEE investments presents a specific challenge, however. To clarify the issue, we decompose the error term in Eq. (1) as follows: $u_{i,t} = \phi\, R_{i,t} + \varepsilon_{i,t}$, where $\varepsilon_{i,t}$ includes all within-neighborhood dynamics that we do not control for, and $R_{i,t}$ represents projected savings imputable to retrofit investments not generating certificates in neighborhood i in year t. The second stage equation thus writes:

$$Y_{i,t} = \beta^{IV}\, \widehat{X}_{i,t} + \delta_2\, \mathbf{W}_{i,t} + \lambda_2\, \mathbf{D}_{i,t} + \alpha_i + \gamma_t + \phi R_{i,t} + \varepsilon_{i,t}$$

where $\widehat{X}_{i,t}$ is the fitted value of $X_{i,t}$. The exclusion restriction requires

$$\mathrm{Cov}(Z_{i,t}, \varepsilon_{i,t}) + \mathrm{Cov}(Z_{i,t}, \phi\, R_{i,t}) = 0. \tag{2}$$

The above argument asserts that temperatures from prior years do not affect current changes in energy use through any channel other than energy retrofits, regardless of whether the investment is subsidized by the program so that $\mathrm{Cov}(Z_{i,t}, \varepsilon_{i,t}) = 0$. However, there is no reason to doubt that past temperature shocks also influence non-CEE energy efficiency investments, leading to $\mathrm{Cov}(Z_{i,t}, R_{i,t}) \neq 0$. As a results, the condition in Eq. (2) no longer holds, resulting in a biased estimate of the coefficient β^{IV}. The good news is that we can sign the bias. Assuming that the control variables are uncorrelated with the instrument, straightforward calculations give:

$$\beta^{IV} = \beta + \phi\, \frac{\mathrm{Cov}(R_{i,t}, Z_{i,t})}{\mathrm{Cov}(X_{i,t}, Z_{i,t})}$$

Focusing on the final term of this expression, it is clear that $\phi < 0$ because non-CEE investments naturally reduce energy consumption. Furthermore, since the first-stage effect of the instrument on CEE-funded investments is not driven by any unique characteristic of the CEE program, it can be extended to all types of energy efficiency retrofits. It follows that $\mathrm{Cov}(R_{i,t}, Z_{i,t})$ and $\mathrm{Cov}(X_{i,t}, Z_{i,t})$ exhibit the same sign. These two statements implies a negative bias: $|\beta^{IV}| > |\beta|$. β^{IV} can be interpreted as a lower-bound estimate of the program's impact on energy use.[13]

13. This issue is common in many studies in which non-subsidized investments are typically unobserved. Christensen et al. (2023) mitigate it by comparing retrofitted homes to not-yet-retrofitted homes, which addresses the concern

6 Econometric results

6.1 Main estimation

We estimate both the OLS and IV regressions for savings, clustering standard errors at the neighborhood level to account for potential within-neighborhood correlation patterns.

Table 3: Regression results for the effect of projected savings on energy use

	OLS-FE	2SLS-FE
Expected Savings	−0.016	
	(0.019)	
Fitted Expected Savings		−0.307***
		(0.087)
Nb. gas heated units	2910.448***	3146.580***
	(284.794)	(298.544)
Nb. elec. heated units	2733.576***	2913.011***
	(287.188)	(292.315)
HDD	2638.098***	2815.555***
	(144.700)	(137.045)
CDD	1168.067*	1723.151***
	(484.657)	(476.158)
Num.Obs.	18 607	18 607
R2 Adj.	0.984	0.983
R2 Within	0.047	0.022
F-test (1st stage)		823.701

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Clustered standard errors at the neighborhood level.

The two columns of Table 3 present the estimated effects of projected savings. The coefficients represent the impact of one kWh of projected savings at time t on energy use in the same year. In column (1), the results indicate that an additional kWh of projected savings at t has no significant

under the assumption that households renovate at most once over the study period.

effect on energy use. However, this observed correlation is endogenous due to the potential selection effects among retrofitting households influenced by within-municipality dynamics, as discussed in Section 5. This endogeneity is further exacerbated by the influence of non-CEE investments.

Column (2) presents the results of the IV estimation, which is based on the framework outlined in Section 5. Compared to the OLS estimate, the effect is negative and significant at all conventional levels. This suggests that failing to account for potential confounders leads to an underestimation of the impact of CEE retrofits on energy use. The discrepancy between $\widehat{\beta}^{OLS}$ and $\widehat{\beta}^{IV}$ likely stems from pre-existing upward trends in energy use within neighborhoods that implement more retrofits.

We estimate that one kWh of projected savings achieved through a CEE operation reduces actual energy use by no more than 0.307 kWh. This represents an overestimation of engineering predictions by at least 69%. The effect is statistically significant at the 1% level and supported by a strong first-stage F-statistic (824), well above the threshold suggested by Staiger and Stock (1997).

The estimated effect should be interpreted as a Local Average Treatment Effect (LATE), similarly to Fowlie, Greenstone, and Wolfram (2018), where compliers are those responding to the financial encouragement. External validity is a priori strong, since temperature is a general determinant of heating and cooling needs that affects the entire population in the same direction. We estimate the impact of certificates rather than specific technologies, which implies a less heterogeneous treatment effect. Estimated effects are similar across subgroups (to be tested formally). The large first-stage F-statistic indicates a large share of compliers.

We run the same IV regression separately for electricity and natural gas. Table 4, column (4), shows that electricity use remains unaffected by additional savings. The effect measured in our main specification in Table 3 is thus mostly triggered by a decrease in natural gas consumption in column (2). This outcome is straightforward to explain: over 80% of the savings are attributed to insulation works in dwellings that mostly do not use electricity for heating. Furthermore, the installation of air-to-water or water-to-water heat pumps – frequently associated with a switch from natural gas to electricity for heating – account for less than 5% of the total savings (see Table 2). Gas-heated houses tend to be older and less insulated, so retrofits yield larger efficiency gains. This also helps explain their higher renovation rates, as such investments are relatively more cost-effective.

Table 4: Effect of projected savings on electricity and gas

	Natural Gas		Electricity	
	OLS-FE	2SLS-FE	OLS-FE	2SLS-FE
Expected Savings	−0.032+		0.029***	
	(0.017)		(0.008)	
Fitted Expected Savings		−0.243***		−0.045
		(0.066)		(0.050)
Nb. gas heated units	1786.342***	1955.295***	1180.958***	1241.064***
	(214.289)	(223.684)	(118.799)	(131.425)
Nb. elec. heated units	1563.627***	1703.371***	1197.699***	1243.373***
	(232.398)	(240.612)	(127.165)	(125.104)
HDD	2384.848***	2516.889***	67.441	112.612*
	(102.505)	(101.334)	(60.700)	(55.339)
CDD	225.415	664.762	959.274***	1100.568***
	(438.419)	(416.568)	(184.509)	(208.141)
Num.Obs.	18 376	18 376	18 607	18 607
R2 Adj.	0.979	0.978	0.988	0.988
R2 Within	0.043	0.024	0.030	0.019
F-test (1st stage)		728.751		823.701

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Clustered standard errors at the neighborhood level.

6.2 Robustness checks

We perform several robustness checks for our IV specification. First, we address potential estimation bias arising from households transitioning away from residential heating oil or liquid gas. While our analysis focuses on neighborhoods with the lowest reliance on these fuels, some scope for such behavior remains. To mitigate this, we exclude municipalities where the share of housing units using heating oil (or liquid gas) changed by more than 1 percentage point between 2018 and 2021. The estimation results for these sub-samples are shown in columns (1) and (2) of Table 5. Although the precision of the coefficient on fitted savings slightly decreases, it remains significant at the 5% level for both fuel sources. The coefficients are of a similar magnitude to those in the main regression, suggesting that at best, only 30% of the projected savings are realized.

Table 5: Robustness checks for the effect of savings on energy use

	Fuel Oil	Liq. Gas	SEM 1km	SEM 5km
Fitted Expected Savings	−0.287**	−0.270**	−0.307**	−0.307+
	(0.093)	(0.086)	(0.103)	(0.167)
Nb. gas heated units	3224.636***	3130.840***	3146.580***	3146.580***
	(334.124)	(300.753)	(301.686)	(350.869)
Nb. elec. heated units	2990.207***	2868.842***	2913.011***	2913.011***
	(323.008)	(293.460)	(299.884)	(300.467)
HDD	2807.033***	2795.107***	2815.555***	2815.555***
	(152.164)	(137.814)	(232.480)	(326.949)
CDD	2014.367***	1584.009***	1723.151**	1723.151+
	(491.816)	(461.620)	(577.210)	(907.097)
Num.Obs.	15 869	18 231	18 607	18 607
R2 Adj.	0.984	0.984	0.983	0.983
R2 Within	0.024	0.028	0.022	0.022
F-test (1st stage)	691.056	788.355	823.701	823.701

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Clustered standard errors at the neighborhood level (1-2) / Spatial Error Model (3-4).

Second, we examine the potential influence of spatial dependence between individual observations. While clustered standard errors account for within-neighborhoods, cross-period correlation, a spatial-error model (SEM) explicitly addresses contemporaneous cross-neighborhoods correlations. In this context, it is plausible that the error term $u_{i,t}$ and $u_{j,t}$ for two adjacent municipalities i and j are not perfectly orthogonal. This may arise because unobserved non-CEE retrofitting investments $R_{i,t}$ and $R_{j,t}$ are influenced by local supply and demand shocks that span across neighborhoods boundaries. To address this, we apply the variance-covariance matrix estimator proposed by Conley (1999), which incorporates the relative proximity of adjacent municipalities using a defined threshold. The error term $u_{i,t}$ of our second stage equation is thus a function of contemporaneous adjacent neighborhoods error terms $u_{j,t}$ and $\vartheta_{i,t} \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$:

$$u_{i,t} = \xi \sum_{i \neq j} \omega_{ij} u_{j,t} + \vartheta_{i,t} \tag{3}$$

We present results for the 1 km and 5 km thresholds in columns (3) and (4) of Table 5. The only deviation from the baseline estimation lies in the computation of standard errors, which are no longer clustered at the municipality level. Nonetheless, the p-value for the effect of contemporaneous savings remains below 5% for the first estimation, and 10% for the second. While this test has limitations – since the spatial diffusion perimeter likely varies across observations, making the optimal radius an empirical question – the consistent significance of the effect of CEE-funded retrofits under this spatial autoregressive specification provides further support for our estimate.

Third, our reference IV model identifies a local average treatment effect (LATE), that is, the impact of certificates on the population of neighborhoods whose investment behavior responds to past weather shocks. To probe the robustness of this estimate, we re-estimate the model using alternative instruments that shift the complier population. We start with the square of lagged HDDs in column (1), Table 6, which amplifies the signal from extreme cold episodes of $t-1$ and thus isolates the effect for households whose investment decisions are particularly sensitive to severe winters. In column (2) of Table 6, we instrument with HDDs lagged two periods rather than one, capturing households that exhibit a longer delay between weather exposure and investment action. Column (3) combines both logics. Consistency of the treatment effect estimates across these alternative instruments provides reassurance that our findings are not specific to the particular complier group defined by the baseline instrument.

Table 6: Alternative Instrument Specifications

	HDD (t-1)²	HDD (t-2)	HDD (t-2)²
Fitted Expected Savings	−0.317***	−0.300***	−0.302***
	(0.090)	(0.086)	(0.090)
Nb. gas heated units	3155.290***	3140.954***	3143.041***
	(299.535)	(298.002)	(298.704)
Nb. elec. heated units	2919.630***	2908.736***	2910.322***
	(292.653)	(292.224)	(292.403)
HDD	2822.101***	2811.327***	2812.895***
	(137.744)	(137.066)	(137.386)
CDD	1743.625***	1709.926***	1714.831***
	(478.847)	(478.272)	(480.518)
Num.Obs.	18 607	18 607	18 607
R2 Adj.	0.983	0.984	0.984
R2 Within	0.020	0.023	0.023
F-test (1st stage)	828.049	822.835	827.986

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Clustered standard errors at the neighborhood level.

Finally, we conduct two robustness checks addressing sample selection and the modelling choice of the analysis. We run the IV estimation for electricity, natural gas, and the sum of both, varying the exclusion threshold from 85 to 95% of electricity- and natural gas-heated residents (Figure B1 in Appendix B). The choice of the 90% baseline threshold is guided by the within-R^2 reported alongside each estimate: at the extremes of the range the within-R^2 turns negative, indicating that the model no longer improves upon a neighbourhood fixed-effect mean alone. This deterioration reflects sample contamination at low thresholds (attenuation from heterogeneous treatment exposure) and thin samples at high thresholds (N falling from 7,676 to 1,598). The 90% threshold sits in the interior of the range where the within-R^2 is positive and stable, point estimates are consistent, and the sample supports credible inference. Across all thresholds, the electricity coefficient remains negative and highly significant, ranging from −0.27 to −0.42 around the baseline of −0.307, while the natural gas coefficients remain small and insignificant, confirming the robustness of the baseline results.

Second, we assess whether the TWFE estimator assigns problematic negative weights to some ATTs following Chaisemartin and d'Haultfœuille 2020 (Appendix B). Because this diagnosis targets the OLS moment conditions rather than the IV exclusion restriction, we apply it to the naive OLS-FE specification. Given that our treatment is continuously distributed and ever-increasing, leaving no clean never-treated comparison group, negative weights are an *a priori* concern. The diagnosis finds that negative weights sum to only -0.089 out of a total absolute weight of approximately 1.18, and the minimum sensitivity parameter required to reverse the sign of the OLS estimate across all cells is 0.046, or four times the magnitude of the estimated coefficient. TWFE aggregation bias therefore provides an unlikely explanation for the OLS–IV gap, which we attribute to endogeneity and classical measurement error.

6.3 Discussion

We estimate that realized savings in residential electricity and gas amount to at most 31% of the engineering-projected savings. Two mechanisms can account for this wedge: (i) a gap between engineering projections and actual savings; and (ii) inframarginal subsidization (free-riding), whereby the policy finances retrofits that would have occurred anyway. For this reason, our estimate is not directly comparable to those available in the study by Fowlie, Greenstone, and Wolfram (2018) or Christensen et al. (2023), who evaluate the impact of the renovations themselves, whereas we identify the impact of the policy designed to induce them.

The existence of such a gap gives a misleading picture of the degree of compliance with national policy objectives. The European Energy Efficiency Directive requires Member States to report annually on the energy savings achieved to meet their national target. Based on the engineering estimates of the CEE, the government reported a compliance rate of 114% for the period 2014-2020 (Ministère de la Transition Écologique 2022). On the contrary, we estimate that realized energy savings in our study sample resulted in a 1.36% annual reduction in energy consumption over 2018-2021, falling short of the -1.5% target set by the Directive (European Commission 2012). Part of the explanation for this energy performance gap likely stems from the program's design. Unlike other foreign energy efficiency schemes (e.g. the Weatherization Assistance Program in the US, the Federal subsidy for efficient buildings in Germany), it neither mandates an energy audit before investments nor includes quality checks upon project completion. More fundamentally, the program adopts a market-based approach that grants obligated parties maximum flexibility in deciding how to achieve their targets. However, in the absence of quality controls, this flexibility may incentivize prioritization of lower-quality, less expensive work or a focus on infra-marginal investments.[14]

7 Estimating the abatement cost of carbon

Among the program's objectives is the reduction of carbon emissions linked to residential energy consumption. This issue is less about electricity consumption, which is mostly decarbonized in France due to nuclear power, than about gas consumption. From this perspective, the fact that most of the savings concern gas is not necessarily a negative outcome. To quantify the impact, we just need to combine the estimates reported in Table 4 with relevant carbon emission factors. According to the French Environmental Agency and the Ministry for Ecology, the average carbon intensity is 79 g tCO_2e/kWh for electricity (ADEME 2020) and 227 g tCO_2e/kWh for natural gas (CGDD 2019). Over the study period, we estimate that the program induced an average annual reduction in carbon emissions of 1.64%.

We build on these results to estimate the cost of reducing carbon emissions achieved through the program. We develop a novel methodology that exploits the tradability of energy savings certificates. The underlying premise is that, assuming a competitive market, certificate prices reflect the marginal cost of achieving energy savings within the CEE scheme. As we will show, the main strength of this revealed-preference approach is its ability to capture all private costs and benefits incurred by obligated firms and households. For instance, the costs faced by energy retailers and their subcontractors when generating certificates include not only the grants required to induce households to undertake energy efficiency investments, but also the transaction and outreach costs associated with identifying and securing these investment opportunities. Interviews with obligated parties suggest that subsidy-related expenditures are particularly substantial. As an illustration, Fowlie, Greenstone, and Wolfram (2015) incorporated an encouragement campaign into their field experiment on the Weatherization Assistance Program in Michigan. The intervention increased participation by 5 percentage points, at a cost of more than USD 1,000 per weatherized household — approximately 20% of the renovation cost — even though the program offered households zero

14. In this way, the program prioritizes cost minimization over the correction of market failures (e.g., inattention, environmental externalities, and split incentives between landlords and tenants) which could yield substantial gains in social welfare as suggested by Allcott, Knittel, and Taubinsky (2015).

out-of-pocket expenses.

Our core strategy for inferring the implied abatement cost of CO_2 is to combine the certificate price with the amount of carbon avoided per certificate. A central challenge arises from a measurement discrepancy: whereas certificate prices reflect *marginal* costs, our empirical estimates pertain to *average* effects on energy consumption. Likewise, available data on the carbon intensity of gas and electricity yield average values. To address this inconsistency, we propose a simple conceptual framework that maps observed market prices to average certificate generation costs. This framework also helps delineate the scope of costs and benefits covered by the approach adopted.

7.1 Conceptual framework

We consider a regulatory setting in which a central authority determines the total number of energy-saving certificates that must be produced over a given compliance period. Let Ω denote this aggregate obligation, which we treat as exogenous, consistent with our objective of analyzing the cost-efficiency properties of the program. A representative obligated firm may fulfill this requirement either by purchasing certificates at a unit price p on the market or by directly incentivizing energy efficiency investments among households.

Let $x_{\max}$, with $x_{\max} > \Omega$, denote the total potential for energy savings across the population. Importantly, this potential corresponds to projected savings, those corresponding to the variable $X_{i,t}$ in the empirical model, except that we consider individual households instead of municipalities. For simplicity, we normalize these projected energy savings such that each household is associated with one unit of potential savings.

The cost of saving energy varies across households. Formally, let $c(x)$ denote the cost of realizing this unit of savings for a household of type x, net of any program subsidy. It thus corresponds to the household's willingness to accept making the investment.[15] Each realized unit generates one certificate.

This willingness to accept investing in one energy saving unit thus encompasses the full spectrum of benefits and costs influencing the retrofit decision. These include monetary savings on energy expenditures, non-monetary gains such as improved comfort (including rebound effects), as well

15. Note that the dependence of c on x does not imply that households base their decision on projected savings or on any particular belief about the energy impact of their choice. However, survey evidence presented by Allcott and Greenstone (2024) in their evaluation of a Wisconsin program suggests that most households take such projections at face value.

as various forms of investment costs—financial, psychological, and practical (e.g., the time and inconvenience associated with renovation works, or the "warm-glow" utility derived from energy conservation). In practice, energy retrofitting is often integrated into broader dwelling renovation projects. The complementarities between the energy-related and non-energy-related components of such projects generate positive spillovers on other aspects of the renovation: $c(x)$ implicitly captures the value of these economies. In addition, c may also include the benefit of subsidies or tax credits that overlap with the CEE program and provide households with additional incentives to undertake retrofits. This is the case in our setting, as households may also benefit from tax credits that reduce the monetary investment cost. These financial incentives will be explicitly accounted for at the calibration stage in order to exclude them into the abatement cost estimation since they constitute a taxpayer-funded transfer.

We assume that household types are ordered such that the private net cost of undertaking the retrofit, denoted $c(x)$, is strictly increasing in x, where $x \in [0, x^{\max}]$. Let x^0 be defined by $c(x^0) = 0$ so that $c(x) \leq 0$ for all $x \in [0, x^0]$. This assumption captures the existence of untapped, privately profitable investments within the population, implying that some subsidy-funded projects may be infra-marginal. The function $c(\cdot)$ is assumed to be linear. This implies that the aggregate cost of inducing energy savings is quadratic in total savings, a property that later allows us to recover average costs from the equilibrium certificate price.

We make two informational assumptions. First, the obligated firm knows the function $c(x)$ but does not observe individual household types. Second, households are initially unaware of the subsidy program and of the possibility of receiving grants; they only become informed when contacted by the firm.[16] As documented in the literature, outreach and persuasion are central components of energy retrofit programs.

Turning to the obligated firm, the cost borne in producing certificates includes two components: (i) the subsidy granted to the investing household, (ii) the cost of identifying and persuading the household to proceed with the retrofit, and converting documentation (e.g., contractor invoices and homeowner attestations of support) into valid certificates once the retrofit is finished. The subsidy amount by unit of projected savings offered to participating households, denoted s, is uniform across recipients, consistent with the firm's inability to observe individual household types.[17] The second

16. This assumption is also consistent with situations in which households only become aware of retrofit opportunities or associated costs upon engaging with the obligated firm.

17. We could assume a more flexible subsidy scheme where the firm would learn the individual type when exchanging

component, the effort required to induce and monitor the participation of a household, is captured by a function $\gamma(x)$ with $\gamma'(x) > 0$. That is, the more reluctant a household is to invest, the higher the cost of convincing it. Again, we assume that $\gamma(.)$ is linear.

Last, the process by which the firm meets its obligation unfolds as follows:

1. The firm determines the share α of its obligation Ω to be met through direct subsidization, with the remainder satisfied via certificate purchases on the market.

2. The firm sets the level of the subsidy s so as to induce a volume of projected savings equal to $\alpha\Omega$ and randomly selects a sample of households to whom the subsidy is offered.

3. Each selected household decides whether to accept the offer. Upon acceptance, the retrofit is carried out, and the firm receives the corresponding number of certificates from the regulator.

4. The firm purchases the remaining certificates needed to meet its total obligation.

In the third stage, a contacted household of type x will accept the subsidy if the net cost of the retrofit is non-positive, i.e., if $c(x) \leq s$. This defines a threshold type $x(s)$ such that all contacted households with $x \leq x(s)$ accept the offer. By construction, the threshold satisfies:

$$c(x(s)) = s.$$

In the previous stage, the firm chooses the subsidy level s to minimize its total cost, subject to the constraint that the number of accepted retrofits equals the obligation level Ω. Cost minimization is straightforward in this context, as the uniform subsidy leads to the self-selection of the lowest-cost households. Furthermore, within a random sample drawn from the overall population of mass $x^{\max}$, the probability that a given household accepts the subsidy offer is $x(s)/x^{\max}$. Compliance with the obligation then requires contacting a mass $[x^{max} \times \Omega]/x(s^*)$ of households.

Last, under the assumption of a competitive certificate market, the equilibrium price p^* is equal to the marginal cost of savings: $p^* = \gamma(x^*) + s^*$. We collect the equilibrium conditions in a lemma.

Lemma 1. *The obligation is fulfilled by*

1. *setting a subsidy s^* such that $c(x(s^*)) = s^*$*

with the household. The uniformity assumption is however realistic as illustrated by web scraped data (Aja and Giraudet 2025).

2. offering it to a random household sample of mass $\Omega \times x_{max}/x(s^*)$.

The resulting certificate price is given by $p^* = \gamma(x(s^*)) + s^*$ and the quantity of certificates produced is equal to $\Omega = [x(s^*)]^2 / x_{max}$.

We can now derive the average cost of generating a certificate. We begin by considering the costs borne by households. Importantly, part of the energy savings induced by the CEE program is achieved at negative cost—that is, for all households of type x such that $x \leq x_0$. Since these households would have undertaken the investment even in the absence of the program, their cost is recorded as zero in our calculation. This distinction is crucial: we are evaluating the cost of the policy, not the cost of the investment itself. The net average cost to households, after subtracting the subsidy, is therefore given by:

$$\left(\int_{x^0}^{x(s^*)} \frac{1}{\Omega x_{max}} c(x) dx \right) - s^* \tag{4}$$

The average cost on the firm's side is

$$\left(\int_{0}^{x(s^*)} \frac{1}{\Omega x_{max}} \gamma(x) dx \right) + s^* \tag{5}$$

Note that we sum the marginal cost starting from 0 rather than from x_0, since generating certificates from inframarginal projects still entails non-negative administrative costs.

The last step of the analysis then consists in relating these costs to the certificate price p^*. From the linearity assumptions of $c(.)$ and $\gamma(.)$ follow that

Proposition 1. *The average cost jointly borne by the obligated firm and the households for achieving one unit of projected savings through the program is equal to $\frac{p^*}{2}$.*

Proof. From the lemma, we have $c(x(s^*)) = s^*$. Since $c(x_0) = 0$ and $c(\cdot)$ is linear, the household's cost, as given by Eq. (4), is equal to $-\frac{s^*}{2}$. Moreover, we know that $p^* = \gamma(x(s^*)) + s^*$, and since $\gamma(\cdot)$ is linear and $c(x) \geq 0$, the firm's cost, as given by Eq. (5), is $\frac{p^* - s^*}{2} + s^*$. Summing both costs, we obtain $\frac{p^*}{2}$. □

This result naturally relies on assumptions. Two are particularly critical: first, the assumption that the marginal cost functions $c(.)$ and $\gamma(.)$, are linear; and second, that the subsidy enters household surplus linearly, implying the absence of income effects. Under these conditions, the subsidy functions purely as a transfer from firms to households.

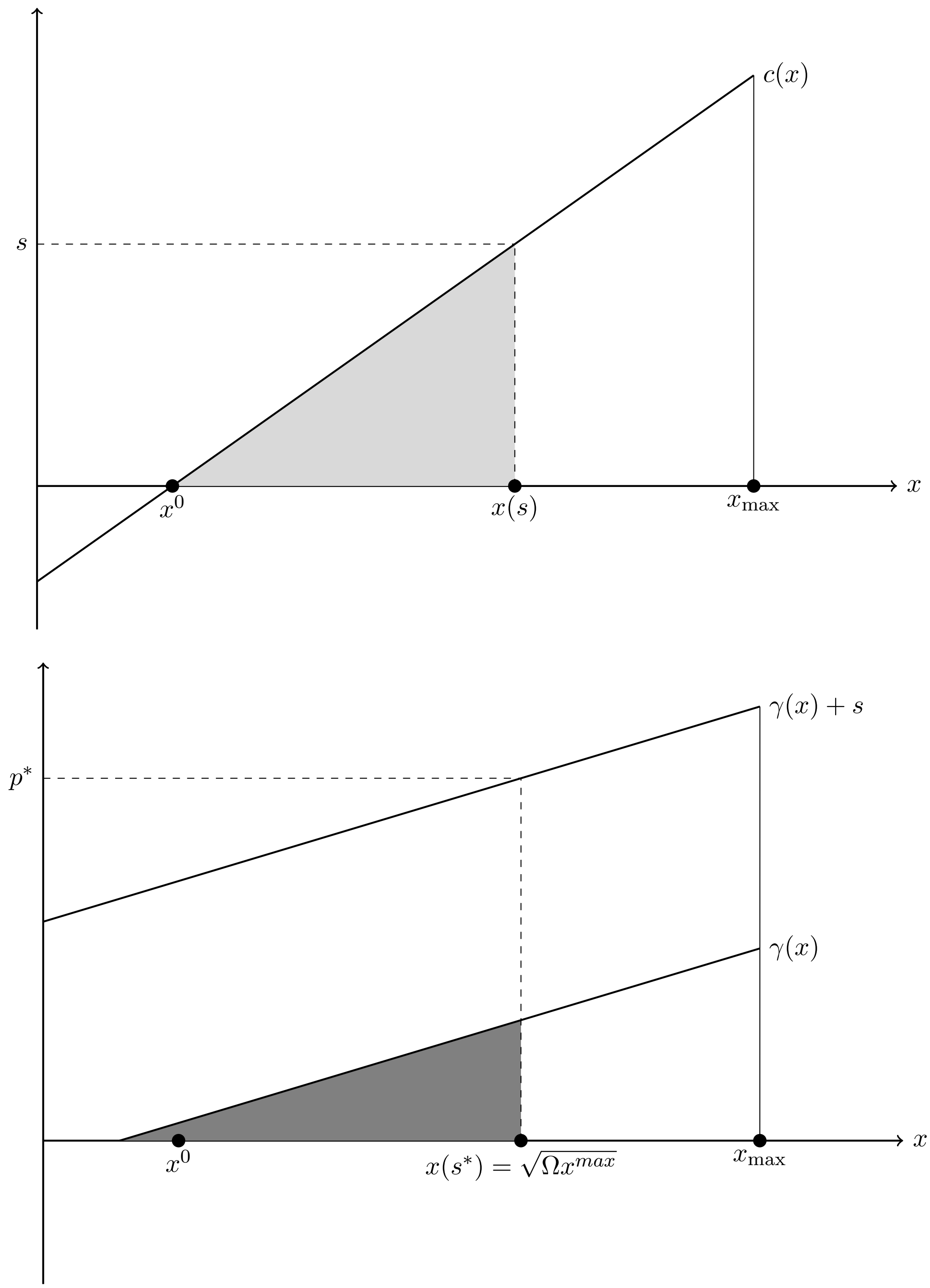


Figure 1: Household response to the CEE subsidy and certificate market equuilibrium

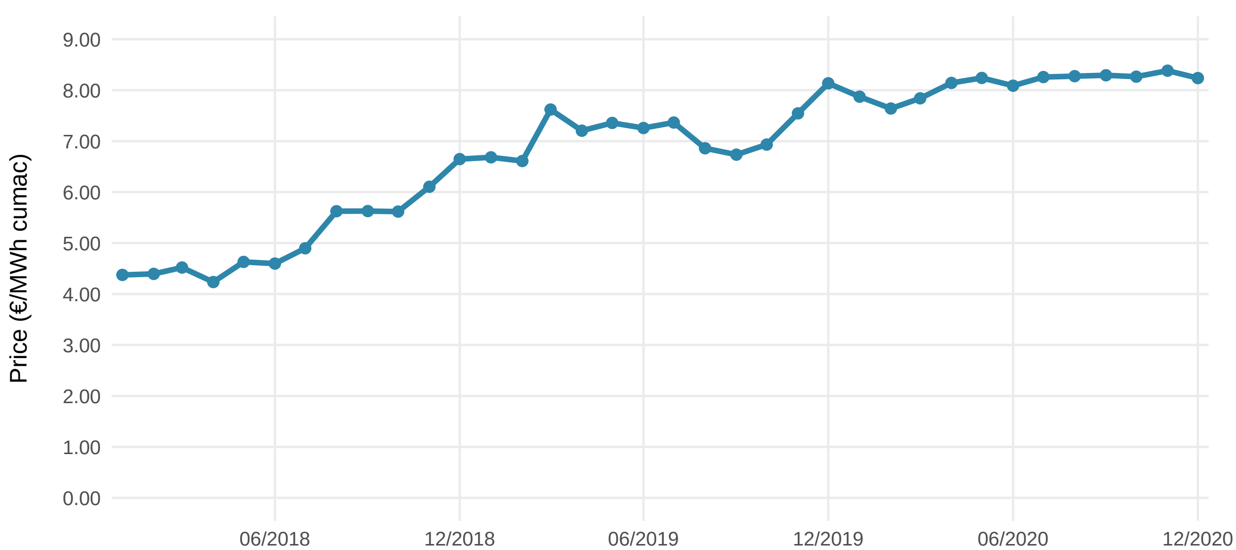
Notes: The household response to a uniform subsidy s is shown in the upper panel, with the grey area depicting the corresponding household loss excluding the subsidy. Note that this does not include the positive surplus of infra-marginal households (those with type $x \leq x^0$). The lower panel displays the resulting certificate market equilibrium with an Ω. The dark grey area represents the encouragement cost borne by the obligated firm to induce the necessary investments, excluding grants. See text for details.

7.2 Calibration and results

Relying on Proposition 1, we are in position to calculate the average cost per ton of CO_2e avoided. Let us first specify the certificate price used in the calculation. This choice is not straightforward, as the price varied between 2018 and 2020. As shown by Figure 2, it increased significantly during most of the study period.

Such a pricing pattern is characteristic of the scheme: each compliance period is associated with an increase in the obligation, which raises the marginal cost of compliance. This upward trend was particularly pronounced during the study period, as the fourth compliance period (2018–2021) involved a sharp increase in the obligation level. In this context, this led to a lag in stakeholders' responses to the new market conditions, reflecting a transitional phase in which economic agents revised their understanding of the new market parameters.

Figure 2: Evolution of the certificate spot price, 2018-2020



Notes: Certificates associated with each project are valued using the spot market price on the contract signature date (Emmy 2023). We rely on official records of savings with bonus in this calculation, because the value of each work is defined by the (annualized) amount of certificates it delivers, including bonuses. Certificates are divided into two categories: standard certificates and so-called "Précarité" certificates, which are issued for investments in dwellings occupied by fuel-poor households. Since early 2018, the prices of both categories have converged, justifying the use of the average price in the graph and subsequent analysis. The reasons for this convergence are detailed by Darmais, Glachant, and Kahn (2024).

Consistent with our theoretical framework, the relevant price is the one influencing the investment decision, that is, the price prevailing when the decision to initiate the works is made. This price reflects the expectations of obligated parties at the moment of investment and therefore serves as the appropriate reference point for the cost of operations initiated at that time.

Second, we derive an expression for the abatement cost that is suitable for calibration using our econometric estimates based on municipality-level data. To do so, we characterize the implications of the theoretical framework developed in the previous section at this level of aggregation. Let x_{imt} denote the number of certificates generated in municipality i from works completed in month m of year t, and let β^g_{imt} and β^e_{imt} be the corresponding average effects on gas and electricity consumption. These effects may vary across municipalities and over time due to heterogeneity in contractor quality or in the composition of retrofits undertaken. Let μ_g and μ_e denote the carbon content per kWh of gas and electricity, respectively, which we assume to be constant over the study period because national trends are absorbed by year fixed effects.

Under these assumptions, Proposition 1 implies that the carbon abated and total cost associated with these certificates in municipality i, month m, and year t are:

$$x_{imt} \times (\beta^g_{imt}\, \mu^g + \beta^e_{imt}\, \mu^e) \quad \text{and} \quad \frac{x_{imt} \times p_{mt}}{2},$$

respectively. Aggregating these costs and abatement quantities across municipalities and over the entire study period, we obtain the average abatement cost of carbon:

$$\frac{1}{2}\left(\sum_{imt} x_{imt} \times p_{mt}\right)\left(\sum_{imt} x_{imt} \times (\hat{\beta}_g \times \mu_g + \hat{\beta}_e \times \mu_e)\right)^{-1} \tag{6}$$

where $\hat{\beta}_g$ and $\hat{\beta}_e$ are the averages of β^g_{imt} and β^e_{imt}, respectively, weighted by certificate quantities x_{it} and estimated in the empirical analysis above. The resulting average cost of carbon abatement through works supported by the CEE is €103/tCO_2e.

Moreover, some renovations were cross-subsidized through a tax credit over the period under study. To quantify the magnitude of this transfer between investing households and taxpayers, we rely on official estimates of overlapping support provided by the French government (ONRE 2022). These data enable us to compute that a share $\pi = 20.74\%$ of retrofit projects were also supported by the tax credit, with an average subsidy of $\sigma =$ €24.5 per MWh of projected energy savings.[18] The term to be added to Eq. (4) to obtain the average household surplus net of this transfer is given by:

18. Details on the underlying assumptions are provided in Appendix C.

$$\pi\,\sigma \times \left(\hat{\beta}_g \times \mu_g + \hat{\beta}_e \times \mu_e\right)^{-1} \tag{7}$$

We are now able to compute the average abatement cost, which is the sum of the expressions in Eq. (6) and (7), using the estimates $\hat{\beta}_g$ and $\hat{\beta}_e$ in Table 4, emission factors from the French environmental agency and the Ministry for Ecology, and monthly certificate price data. This yields an estimate of €179/tCO_2e.

This estimate per ton is significantly higher than the official French carbon value set for 2020, which was €90/tCO_2e. This reference value does not correspond to the social cost of carbon, but rather to the shadow cost of achieving the national abatement target included in France's Nationally Determined Contribution (NDC). It has since been substantially revised upward to reflect the updated target of a 55% reduction in greenhouse gas emissions by 2030 and now amounts to €250/tCO_2e. Relative to this benchmark, the average energy retrofit measures included in the study sample meet the cost-effectiveness criterion within the French climate policy mix.

Last, it mostly falls below the other empirical estimates available in the literature. If we only consider the cost arising from the energy efficiency obligation itself, it stands below that of Fowlie, Greenstone, and Wolfram (2018), who estimate it at USD 200 per ton in the context of the Weatherization Assistance Program. The results are however not directly comparable. First, the scope of the costs and benefits are different. Fowlie and coauthors account for the impact of the energy use reductions on the emission of local pollutants and the private benefits associated of increasing indoor temperatures after retrofit (i.e., the rebound effect), the other non-monetary costs benefits as well as the encouragement cost. Moreover, they include in their estimation the costs and benefits of the inframarginal renovations while they are set to zero in our analysis. Second, our estimate covers all non-monetary costs and benefits that are priced by the market for certificates, including the discomfort incurred during installation, comfort gains, and the transaction and commercialization costs. To the best of our knowledge, we are the first to estimate such a comprehensive cost of abatement, specifically for energy retrofit investments.

It is important to emphasize that $c(x)$ reflects households' ex ante payoffs based on the information set available to households before the investment. It does not correspond to ex-post realizations, which may diverge from expectations owing to systematic misperceptions of costs and benefits. This distinction is salient in the energy efficiency literature, where these misperceptions are frequently cited as a source of the discrepancy between the investments households actually undertake and those that would appear privately profitable, a phenomenon commonly referred to as the "energy efficiency gap".

7.3 Effect of energy price increases through the CEE Program

The analysis conducted thus far provides only a partial assessment of the program's carbon impact. As outlined in introduction, the cost borne by energy retailers to generate certificates is ultimately transmitted to end-user energy prices. This pass-through mechanism induces all households, rather than solely those undertaking energy efficiency investments, to reduce their energy consumption. In this subsection, we seek to quantify this indirect effect, which amplifies the overall carbon impact of the program, and the associated abatement cost. The analysis is conducted within the empirical scope of our study, namely the municipalities included in our sample and their residential consumption of gas and electricity.

To compute the energy user welfare loss due to the energy price increase, we rely on a standard demand framework. Let $m_{f,t}$ denote the price of fuel f in year t with $f \in \{e, g\}$ and $m^0_{f,t}$ the counterfactual price in the absence of the program. Assuming (locally) linear demand functions for gas and electricity, the total consumer welfare loss due to the program-induced increase in fuel prices is

$$\frac{1}{2} \times \sum_{f,i,t} \left(m_{f,t} - m^0_{f,t}\right) \left(Y^0_{f,i,t} - Y_{f,i,t}\right), \tag{8}$$

This Harberger-type distorsion corresponds to the hatched area in Figure 3.

By applying appropriate emission factors to the reduced consumption of gas and electricity corresponding quantity of carbon abated is given by:

$$\sum_{f,i,t} \mu_f \times \left(Y^0_{f,i,t} - Y_{f,i,t}\right),$$

so that the average abatement cost of carbon is

$$C := \frac{1}{2} \times \frac{\sum_{f,i,t} \left(m_{f,t} - m^0_{f,t}\right) \left(Y^0_{f,i,t} - Y_{f,i,t}\right)}{\sum_{f,i,t} \left(Y^0_{f,i,t} - Y_{f,i,t}\right) \times \mu_f} \tag{9}$$

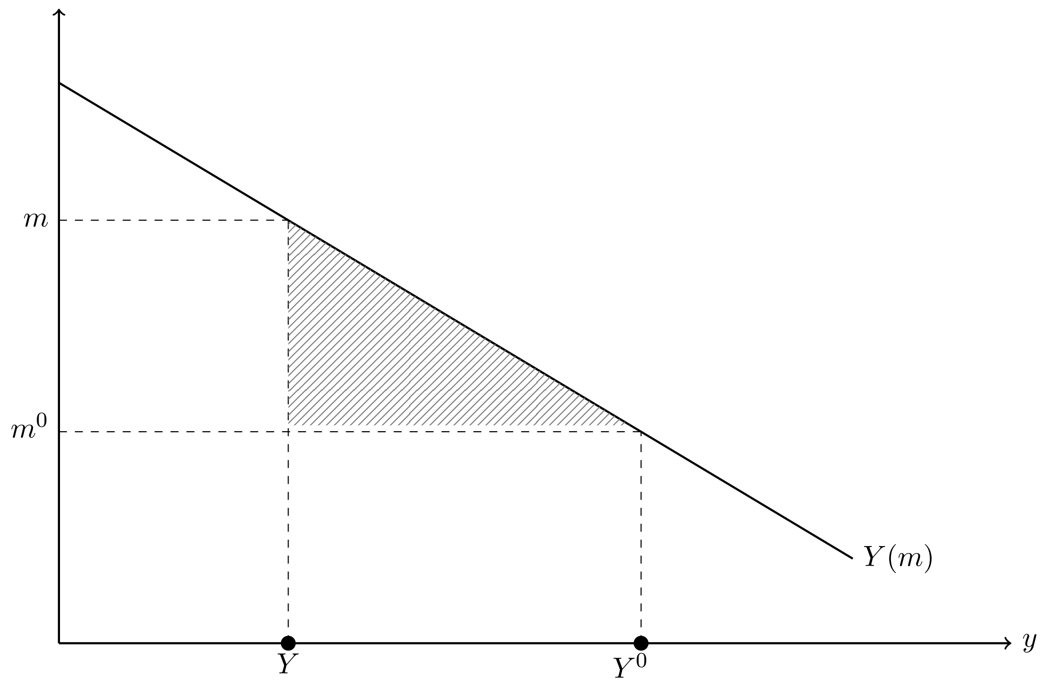


Figure 3: Impact of the program on energy demand

Notes: The graph shows the effect of the program-induced increase in energy prices on energy demand and the resulting decline in consumer surplus (the hatched area).

However, this expression does not permit the direct computation of abatement costs, since counterfactual fuel prices $m^0_{f,t}$ and fuel consumption levels $Y^0_{f,i,t}$ are not observed. To address this limitation, we rely on external estimates of pass-through rates and energy-demand elasticities from the existing literature.

Relying on administrative data, Darmais, Glachant, and Kahn (2024) estimate the difference $m_{f,t} - m^0_{f,t}$ for the year 2019. They find that the program led to a 4.61% increase in residential gas prices and a 3.16% increase in electricity prices relative to the counterfactual. These rates cannot, however, be applied uniformly over the entire study period. As shown in Figure 2, certificate prices exhibit substantial variation over time, implying corresponding changes in their impact on energy prices. We therefore infer program-induced fuel price increases for 2018, 2020 and 2021 by scaling the 2019 estimates according to the relative changes in certificate prices observed in those years. Details are provided in Appendix D.

Ultimately, we can compute the price difference using the following expression:

$$m_{f,t} - m^0_{f,t} = m_{f,t}\left(\frac{\delta_{f,t}}{1+\delta_{f,t}}\right), \tag{10}$$

where $\delta_{f,t}$ denotes the program-induced price increase rate of fuel f in year t.

Turning next to the estimation of the difference $Y^0_{f,i,t} - Y_{f,i,t}$, we rely on demand elasticity estimates for the French residential sector from Douenne (2020). The paper provides separate elasticities for 5 urbanization categories, building on the classification defined by the INSEE. We apply these estimates and find a weighted average demand elasticity of -0.0784[19]. A limitation of these estimate is that it does not distinguish between gas and electricity consumption. We therefore assume identical elasticities across fuels, an assumption supported by Labandeira, Labeaga, and López-Otero (2017), whose meta-analysis finds no statistically significant difference between electricity and gas price elasticities.

Denoting this price elasticity by ϵ, we have by definition

$$-\epsilon \;=\; \left(\frac{Y^0_{f,i,t} - Y_{f,i,t}}{Y^0_{f,i,t}}\right)\left(\frac{m_{f,t} - m^0_{f,t}}{m^0_{f,t}}\right)^{-1}. \tag{11}$$

We then substitute Eqs. (10) and (11) into Equation (9) and rearrange leading to a computable expression for the average carbon abatement cost:

$$C = \frac{1}{2}\left(\sum_{f,i,t} \frac{m_{f,t}\, Y_{f,i,t}\, \left(\delta_{f,t}\right)^2}{(1+\delta_{f,t})\,(1+\epsilon\,\delta_{f,t})}\right)\left(\sum_{f,i,t} \frac{\mu_f Y_{f,i,t}}{1+\epsilon\delta_{f,t}}\delta_{f,t}\right)^{-1} \tag{12}$$

Finally, we parameterize the model by combining demand elasticity estimates and program-induced price increase rates from the literature with observed certificate prices and energy consumption quantities in our dataset. The resulting average cost of carbon abatement induced by the increase in energy prices amounts to €7.76/tCO_2e.

We compute the overall average cost of carbon abatement as the sum of the subsidy and price effects of the policy, weighted by their respective carbon abatement. According to the denominators in Eq. (6) and (12), energy efficiency investments supported by the policy abated 695,435 tons of CO_2e, while the induced increase in energy prices lead to an additional abatement of 1,889,442 tons, three times more. The resulting overall cost is equal to €55/tCO_2e.

19. The estimated elasticity are reported in Table A1 in appendix.

8 Conclusion

Relying on a new administrative dataset recording all CEE-supported energy efficiency retrofits in the French residential sector between 2018 and 2021, we investigate their impact on gas and electricity consumption in 4,671 urban municipalities. We compare our econometric estimates associated to these operations to measure the wedge between projected and realized savings. We document two primary findings. First, each kWh of savings predicted by engineering models yields in the best case scenario only 0.31 kWh of realized savings. The French CEE is therefore plagued by a minimum 69% energy performance gap. Second, the average cost of carbon abatement through CEE retrofits is equal to €181 per ton of CO_2 equivalent. This estimate however gives a partial view on the program outcome as it ignores its effect on the energy prices. When accounting for this energy price channel, the abatement cost falls to €54/tCO_2e.

We acknowledge several important limitations of the present study. First, we are unable to provide a definitive estimate of the actual energy savings induced by the French program. This limitation mainly stems from our inability to control for contemporaneous energy retrofits not supported by the program. As a result, our estimates should be interpreted as a lower bound on the energy performance gap, and we remain agnostic regarding its underlying determinants. In particular, we do not investigate the two information asymmetries highlighted in the literature: moral hazard arising from information asymmetries between beneficiaries and installers, and behavioral responses by beneficiary households, commonly referred to as the rebound effect.

Second, the estimation of abatement costs relies on a model that rests on several assumptions. While these assumptions differ from those underlying accounting-style calculations—typically concerning investment lifetimes and discount rates—they are also open to debate. Data limitations prevent us from formally testing the robustness of the results derived from this model.

On a broader point of view, one should also keep in mind that energy efficiency works may have other effects than energy reduction and carbon emissions reductions, such as health improvements or unemployment reduction through the creation of green jobs.

References

ADEME. 2020. "Positionnement de l'ADEME sur le calcul du contenu CO_2 de l'électricité, cas du chauffage électrique."

Aja, Pille-Riin and Giraudet, Louis-Gaëtan. 2025. "On the added value of utility-sponsored energy efficiency subsidies: Insights from the French White Certificates program, Phase III."

Allcott, Hunt. 2011. "Consumers' Perceptions and Misperceptions of Energy Costs." *American Economic Review* 101 (3): 98–104. https://doi.org/10.1257/aer.101.3.98.

Allcott, Hunt and Greenstone, Michael. 2024. *Measuring the Welfare Effects of Residential Energy Efficiency Programs.* NBER Working Paper 23386. National Bureau of Economic Research. https://doi.org/10.3386/w23386.

Allcott, Hunt, Knittel, Christopher, and Taubinsky, Dmitry. 2015. "Tagging and Targeting of Energy Efficiency Subsidies." *American Economic Review* 105 (5): 187–91. https://doi.org/10.1257/aer.p20151008.

Allcott, Hunt, Mullainathan, Sendhil, and Taubinsky, Dmitry. 2014. "Energy policy with externalities and internalities." *Journal of Public Economics* 112:72–88. ISSN: 0047-2727. https://doi.org/https://doi.org/10.1016/j.jpubeco.2014.01.004.

Arimura, Toshi H., Newell, Richard G., Palmer, Karen, and Stavins, Robert N. 2012. "Cost-effectiveness of electricity energy efficiency programs." *The Energy Journal* 33 (2): 63–99.

Aroonruengsawat, Anin, Auffhammer, Maximilian, and Sanstad, Alan H. 2012. "The impact of state-level energy efficiency programs on electricity consumption in the United States." *The Energy Journal* 33 (1): 159–198.

Benzarti, Youssef. 2020. "How Taxing Is Tax Filing? Using Revealed Preferences to Estimate Compliance Costs." *American Economic Journal: Economic Policy* 12 (4): 38–57. https://doi.org/10.1257/pol.20180664.

Bernheim, B. Douglas and Taubinsky, Dmitry. 2018. *Behavioral Public Economics.* NBER Working Paper 24828. National Bureau of Economic Research. https://doi.org/10.3386/w24828.

Boomhower, Judson and Davis, Lucas W. 2014. "A credible approach for measuring inframarginal participation in energy efficiency programs." *Journal of Public Economics* 113 (C): 67–79. https://doi.org/10.1016/j.jpubeco.2014.03.009.

Brandon, Alec, Clapp, Christopher M., List, John A., Metcalfe, Robert D., and Price, Michael. 2022. "The Human Perils of Scaling Smart Technologies: Evidence from Field Experiments."

Broc, Jean-Sébastien, Stańczyk, Wojciech, and Reidlinger, Bettin. 2020. *Snapshot of Energy Efficiency Obligation Schemes in Europe.* Report Provisional version.

CASD. 2023. "CEE : Energy saving certificates."

CGDD, Commissariat général au développement durable. 2019. "Les facteurs d'émission de gaz à effet de serre."

Chaisemartin, Clément de and d'Haultfœuille, Xavier. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–2996. https://doi.org/10.1257/aer.20181169.

Chetty, Raj. 2015. "Behavioral Economics and Public Policy: A Pragmatic Perspective." *American Economic Review* 105 (5): 1–33. https://doi.org/10.1257/aer.p20151108.

Christensen, Peter, Francisco, Paul, Myers, Erica, and Souza, Mateus. 2023. "Decomposing the Wedge between Projected and Realized Returns in Energy Efficiency Programs." *The Review of Economics and Statistics* 105 (4): 798–817. ISSN: 0034-6535. https://doi.org/10.1162/rest_a_01087.

Conley, T.G. 1999. "GMM estimation with cross sectional dependence." *Journal of Econometrics* 92 (1): 1–45. ISSN: 0304-4076. https://doi.org/https://doi.org/10.1016/S0304-4076(98)00084-0.

Crampes, Claude and Léautier, Thomas-Olivier. 2020. *White certificates and competition.* TSE Working Papers 20-1167. Toulouse School of Economics (TSE).

Currie, Janet and Musen, Kate. 2025. *Information, Spillovers, or Hassle Costs? Effects of Medicaid Prior Authorization on Preschool Antipsychotic Prescribing.* Working Paper 34369. National Bureau of Economic Research. https://doi.org/10.3386/w34369.

Darmais, Amélie, Glachant, Matthieu, and Kahn, Victor. 2024. "Social equity provisions in energy efficiency obligations: An ex-post analysis of the French program." *Energy Policy* 195:114348. ISSN: 0301-4215. https://doi.org/https://doi.org/10.1016/j.enpol.2024.114348.

Davis, Lucas W., Fuchs, Alan, and Gertler, Paul. 2014. "Cash for Coolers: Evaluating a Large-Scale Appliance Replacement Program in Mexico." *American Economic Journal: Economic Policy* 6 (4): 207–38. https://doi.org/10.1257/pol.6.4.207.

Davis, Lucas W., Martinez, Sebastian, and Taboada, Bibiana. 2020. "How effective is energy-efficient housing? Evidence from a field trial in Mexico." *Journal of Development Economics* 143:102390. ISSN: 0304-3878. https://doi.org/https://doi.org/10.1016/j.jdeveco.2019.102390.

DGEC. 2022. *Bilan de la 4eme periode des CEE, 2018-2021.* Technical report. Direction Générale de l'Energie et du Climat.

Direction générale des Finances publiques. 2022. *Bilan de l'utilisation du crédit d'impôt transition énergétique (CITE).* Rapport administratif. Direction générale des Finances publiques (DG-FiP).

Douenne, Thomas. 2020. "The Vertical and Horizontal Distributive Effects of Energy Taxes: A Case Study of a French Policy." *The Energy Journal* 41 (3): 231–254. https://doi.org/10.5547/01956574.41.3.tdou.

Drake, Coleman, Ryan, Conor, and Dowd, Bryan. 2022. "Sources of inertia in the individual health insurance market." *Journal of Public Economics* 208:104622. ISSN: 0047-2727. https://doi.org/https://doi.org/10.1016/j.jpubeco.2022.104622.

Emmy, Registre National des Certificats d'Economie d'Energie. 2023. "Cotation du kWh cumac."

European Commission. 2012. "Directive 2012/27/EU of the European Parliament and of the Council of 25 October 2012 on energy efficiency, amending Directives 2009/125/EC and 2010/30/EU and repealing Directives 2004/8/EC and 2006/32/EC." *Off Journal of the European Union, E. Parliament, Editor.*

Fowlie, Meredith, Greenstone, Michael, and Wolfram, Catherine. 2015. "Are the Non-monetary Costs of Energy Efficiency Investments Large? Understanding Low Take-Up of a Free Energy Efficiency Program." *American Economic Review* 105 (5): 201–04. https://doi.org/10.1257/aer.p20151011.

———. 2018. "Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program." *The Quarterly Journal of Economics* 133 (3): 1597–1644.

Gerarden, Todd D., Newell, Richard G., and Stavins, Robert N. 2017. "Assessing the Energy-Efficiency Gap." *Journal of Economic Literature* 55 (4): 1486–1525. https://doi.org/10.1257/jel.20161360.

Gillingham, Kenneth and Stock, James H. 2018. "The Cost of Reducing Greenhouse Gas Emissions." *The Journal of Economic Perspectives* 32 (4): 53–72. ISSN: 08953309.

Giraudet, Louis-Gaëtan, Glachant, Matthieu, and Nicolaï, Jean-Philippe. 2020. "Selling and Saving Energy: Energy Efficiency Obligations in Liberalized Energy Markets." *The Energy Journal* 41 (Special I).

Giraudet, Louis-Gaëtan, Houde, Sébastien, and Maher, Joseph. 2018. "Moral Hazard and the Energy Efficiency Gap: Theory and Evidence." *Journal of the Association of Environmental and Resource Economists* 5 (4): 755–790.

Houde, Sebastien, Papineau, Maya, Rivers, Nicholas, and Yassin, Kareman. 2025. *Big Hassle on the Decarbonization Frontier.* Technical report 12326. CESifo Working Paper. https://doi.org/10.2139/ssrn.5908870.

Houde, Sébastien and Myers, Erica. 2019. *Heterogeneous (Mis-) Perceptions of Energy Costs: Implications for Measurement and Policy Design.* NBER Working Paper 25722. https://doi.org/10.3386/w25722.

IEA. 2022. *Energy Efficiency.* Technical report. International Energy Agency.

INSEE. 2018. "Logement en 2018, Recensement de la population - Base infracommunale (IRIS)."

———. 2020. (Base des aires d'attraction des villes 2020. Updated March 19, 2026; accessed May 1, 2026).

Institut national de la statistique et des études économiques (INSEE). 2016. *IRIS — Definition.* https://www.insee.fr/en/metadonnees/definition/c1523. INSEE.

Kotchen, Matthew J. 2017. "Longer-Run Evidence on Whether Building Energy Codes Reduce Residential Energy Consumption." *Journal of the Association of Environmental and Resource Economists* 4 (1): 135–153. https://doi.org/10.1086/689703.

La Nauze, Andrea and Myers, Erica. 2023. *Do Consumers Acquire Information Optimally? Experimental Evidence from Energy Efficiency.* NBER Working Paper 31742. National Bureau of Economic Research. https://doi.org/10.3386/w31742.

Labandeira, Xavier, Labeaga, José M., and López-Otero, Xiral. 2017. "A meta-analysis on the price elasticity of energy demand." *Energy Policy* 102:549–568. ISSN: 0301-4215. https://doi.org/https://doi.org/10.1016/j.enpol.2017.01.002.

Lang, Ghislaine and Lanz, Bruno. 2022. "Climate policy without a price signal: Evidence on the implicit carbon price of energy efficiency in buildings." *Journal of Environmental Economics and Management* 111:102560. ISSN: 0095-0696. https://doi.org/https://doi.org/10.1016/j.jeem.2021.102560.

Levinson, Arik. 2016. "How Much Energy Do Building Energy Codes Save? Evidence from California Houses." *American Economic Review* 106 (10): 2867–94. https://doi.org/10.1257/aer.20150102.

McKinsey. 2009. *Pathways to a low-carbon economy: Version 2 of the global greenhouse gas abatement cost curve.* Technical report. McKinsey Company.

Ministère de la Transition Écologique. 2022. *Rapport de la France en application de l'article 27 du règlement 2018/1999 du Parlement européen et du Conseil du 11 décembre 2018 sur la gouvernance de l'union de l'énergie et de l'action pour le climat – Actualisation 2022* [in French]. Rapport annuel. Ministère de la Transition Écologique et de la Cohésion des Territoires. Accessed May 1, 2026.

Myers, Erica. 2020. "Asymmetric information in residential rental markets: Implications for the energy efficiency gap." *Journal of Public Economics* 190:104251. ISSN: 0047-2727. https://doi.org/https://doi.org/10.1016/j.jpubeco.2020.104251.

ONRE. 2022. *La rénovation énergétique des logements : Bilan des travaux et des aides entre 2016 et 2019.* Technical report. Ministère de la Transition Ecologique (MTE).

Papineau, Maya, Rivers, Nicholas, and Yassin, Kareman. 2025. "Household benefits from energy efficiency retrofits: Implications for net zero housing policy." *Energy Economics* 143:108245. ISSN: 0140-9883. https://doi.org/https://doi.org/10.1016/j.eneco.2025.108245.

Peñasco, Cristina and Anadón, Laura Díaz. 2023. "Assessing the effectiveness of energy efficiency measures in the residential sector gas consumption through dynamic treatment effects: Evidence from England and Wales." *Energy Economics* 117:106435.

Pisany-Ferry, Jean and Mahfouz, Selma. 2023. *Les incidences économiques de l'action pour le climat.* Technical report. France Stratégie.

Rosenow, Jan and Bayer, Edith. 2017. "Costs and benefits of Energy Efficiency Obligations: A review of European programmes." *Energy Policy* 107:53–62. ISSN: 0301-4215. https://doi.org/https://doi.org/10.1016/j.enpol.2017.04.014.

Rosenow, Jan, Cowart, Richard, and Thomas, Samuel. 2019. "Market-based instruments for energy efficiency: a global review." *Energy Efficiency* 12 (5): 1379–1398.

Service des données et études statistiques. 2024. *Les rénovations énergétiques par geste aidées par MaPrimeRénov' en 2023.* Rapport statistique. Ministère de la Transition écologique, SDES.

Staiger, Douglas and Stock, James H. 1997. "Instrumental Variables Regression with Weak Instruments." *Econometrica* 65 (3): 557–586. ISSN: 00129682, 14680262, accessed September 6, 2023.

Vivier, Lucas and Giraudet, Louis-Gaëtan. 2024. "Energy efficiency policy in an n-th best world: Assessing the implementation gap." Https://hal.science/hal-04510798v2.

Zivin, Joshua Graff and Novan, Kevin. 2016. "Upgrading Efficiency and Behavior: Electricity Savings from Residential Weatherization Programs." *The Energy Journal* 37 (4): 1–23. ISSN: 01956574, 19449089.

A Data Appendix

This appendix provides contextual descriptive statistics for the estimation sample and details the construction of the weather variables used in the analysis.

Sample Composition. The estimation sample comprises 4,671 neighborhoods, drawn from the 44,093 neighborhoods included in the full administrative dataset. The restriction follows from the 90% exclusion threshold described in Section 4: only neighborhoods where at least 90% of residential units rely on electricity (respectively natural gas) for space heating are retained, ensuring that aggregate energy consumption at the neighborhood level responds cleanly to the subsidized retrofit operations targeting those heating technologies. The resulting sample is therefore a high-penetration subsample rather than a random draw, and generalizing the estimated treatment effects beyond it requires caution.

Table A1 reports the distribution of neighborhoods across INSEE urbanization categories, alongside the residential energy demand elasticities for France estimated by Douenne 2020 using the same classification. The estimation sample spans the full urbanisation gradient, from dense urban cores to rural areas, though the share of peri-urban and intermediate neighborhoods is proportionally larger than in the national population. This composition is consequential for interpreting the estimated treatment effects: demand elasticities vary substantially across the gradient (Douenne 2020), so the sample-weighted average effect need not coincide with a national average.

Table A1: Neighborhoods by urbanization & demand elasticity for residential energy

Size group	Number of neighborhoods	Elasticity (Douenne 2020)
Paris	982	0.135
Large cities	2,679	−0.082
Medium cities	768	−0.270
Small cities	239	−0.294
Rural	3	−0.339

Source: INSEE, *Aires d'attraction des villes.* Urbanization categories follow the degree-of-urbanization typology defined by INSEE, which classifies municipalities into functional urban areas based on commuting flows and population density (INSEE, Base des aires d'attraction des villes 2020). Demand elasticities for residential energy use in France are taken from Douenne 2020, who adopts the same structure based on urbanization categories.

Table A2: Summary statistics on the 28 standardized energy retrofit operations used in the residential sector

Operation	**Life exp. (years)**	**Mean (MWh/y)**	**Median (MWh/y)**	**Sum 2018–20 (GWh/y)**	**Total (%)**	**Rank**
Roof insulation	30	7.691	6.617	13814.345	31.31	1
Heating / hot water network insulation	20	133.410	49.774	6995.104	15.86	2
Floor insulation	30	10.467	6.317	6800.054	15.41	3
Wall insulation	30	22.092	13.101	6777.157	15.36	4
High EE individual boiler	17	3.932	3.604	2358.890	5.35	5
HP air-water / water-water	17	6.649	6.315	1975.209	4.48	6
Window w/ insulated glazing	24	4.323	2.069	901.950	2.04	7
Independent wood-burning system	13	2.893	3.033	712.905	1.62	8
High EE collective boiler	22	105.662	63.210	679.407	1.54	9
HP air-air	17	3.824	3.169	635.851	1.44	10
Heating network insulation	20	176.878	100.637	432.114	0.98	11
Rooftop insulation	30	44.932	27.647	400.884	0.91	12
Individual biomass boiler	17	8.417	9.200	355.314	0.81	13
High EE collective boiler w/ contract	22	206.124	123.687	337.837	0.77	14
Simple-flow hygro-adjustable CMV	17	6.026	2.948	322.396	0.73	15
Hot water network insulation	20	196.595	108.781	304.525	0.69	16
HP water-heater	17	1.230	1.233	94.991	0.22	17
Deep individual renovation	30	15.097	13.191	58.469	0.13	18
Hybrid individual HP	17	8.727	7.942	47.892	0.11	19
Dual-flow ventilation	17	3.541	3.320	25.456	0.06	20
Deep collective renovation	30	182.000	106.673	24.752	0.06	21
Hybrid hygro-adjustable ventilation	17	39.793	14.985	21.846	0.05	22
Individual solar water heater	20	1.625	1.521	19.939	0.05	23
Low-temperature radiator	35	1.782	0.613	13.572	0.03	24
Insulating closure	24	0.918	0.218	3.125	0.01	25
Low-temperature underfloor heating system	50	1.442	1.309	1.085	0.00	26
Collective solar water heater	22	32.811	16.242	0.623	0.00	27
Collective HP	22	19.658	17.513	0.098	0.00	28

Standardized Retrofit Operations. Table A2 summarises the 28 standardised energy retrofit operation types (*fiches d'opérations standardisées*) eligible under the *Certificats d'Économies d'Énergie* scheme in the residential sector. The table captures variation in the type of intervention (insulation, heating systems, renewable generation), the scale of the associated *cumac* savings in kWh, and the relative frequency of each operation in full sample (see Table 2 for the estimation sample counterpart). Our modelling choice relying on neighbourhood-level treatment variable (expected savings in kWh) allows us to abstract from modelling operation types separately.

Computation of Heating and Cooling Degree Days. Heating Degree Days (HDD) and Cooling Degree Days (CDD) are based on the principle that no heating or cooling is required when the outdoor temperature is comprised in between 17 and 23°C. When the daily mean temperature exceeds 23°C, the difference represents Cooling Degree Days (CDD); when it falls below 17°C, the difference represents Heating Degree Days (HDD).

Formally, for each node n and day d, we compute HDD following the SDES methodology as:

$$HDD_{n,d} = \begin{cases} 17 - T_{n,d}, & \text{if } T_{n,d} < 17°\text{C} \\ 0, & \text{otherwise.} \end{cases}$$

We then aggregate HDD values at the annual level for each node:

$$HDD_{n,y} = \sum_{d=1}^{365} HDD_{n,d},$$

and subsequently compute annual HDD for each iris ($HDD_{i,y}$).

Mapping HDD to the IRIS level is performed according to the following rule:

$$HDD_{i,y} = \begin{cases} HDD_{n,y}, & \text{if } n \subset m \text{ and there is no other node within } m, \\ \frac{1}{k}\sum_{i=1}^{k} HDD_{n_i,y}, & \text{if } \exists\, k > 1 \text{ such that } n_i \subset m, \\ HDD_{n^*,y}, \quad n^* = \arg\min_n \|C_m - n\|, & \text{if no node lies within } m, \end{cases}$$

where C_i denotes the centroid of IRIS i, used to assign HDD values via one-nearest-neighbor matching when no temperature node falls within the IRIS boundaries. $CDD_{n,d}$ and $CDD_{i,y}$ follow the same logic applied to daily mean temperatures exceeding 23°C.

B Robustness Analysis

Sample selection. We run the IV estimation for electricity, natural gas, and the sum of both, varying the exclusion threshold from 85 to 95% of electricity- and natural gas-heated residents. The choice of the 90% exclusion threshold adopted in the baseline is guided by the within-R^2 pattern reported alongside each estimate in Figure B1. At the extremes of the threshold range, particularly below 87% and above 93%, the within-R^2 turns negative for the electricity and combined specifications, indicating that the model fits the within-neighbourhood variation no better than a neighbourhood fixed-effect mean alone. This deterioration likely reflects two distinct sources of sample contamination. At low thresholds, neighbourhoods with a substantial share of non-electric or non-gas heated units are retained, introducing attenuation bias from heterogeneous treatment exposure. At high thresholds, the sample shrinks considerably, from N=7,676 at the 85% threshold down to N=1,598 at 95%, reducing both the precision and the representativeness of the estimates. A negative within-R^2 in a two-way fixed-effects context does not indicate model misspecification *per se*; rather, it signals that the treatment variable contributes no explanatory power beyond the fixed effects for that subsample, which is itself informative about sample quality.[20] The 90% threshold sits at the interior of the range where the within-R^2 is positive and relatively stable, the point estimates are consistent, and the sample remains large enough to support credible inference. We therefore adopt it as our baseline.

Figure B1 displays the results across thresholds. The coefficient on fitted expected savings for natural gas consumption is negative and highly significant across all eleven thresholds, ranging from -0.27 to -0.42, and closely tracks the baseline estimate of -0.307 (Table 3). The combined gas-and-electricity estimates are similarly stable, suggesting that the natural gas margin drives the aggregate result. By contrast, the electricity coefficients are small in magnitude, statistically insignificant at conventional levels for most thresholds, and centred near zero, consistent with the baseline.

20. Formally, within-R^2 can be negative when the included regressors increase the residual sum of squares relative to the within-demeaned intercept-only model, a well-known artifact of TWFE with sparse treatment variation.

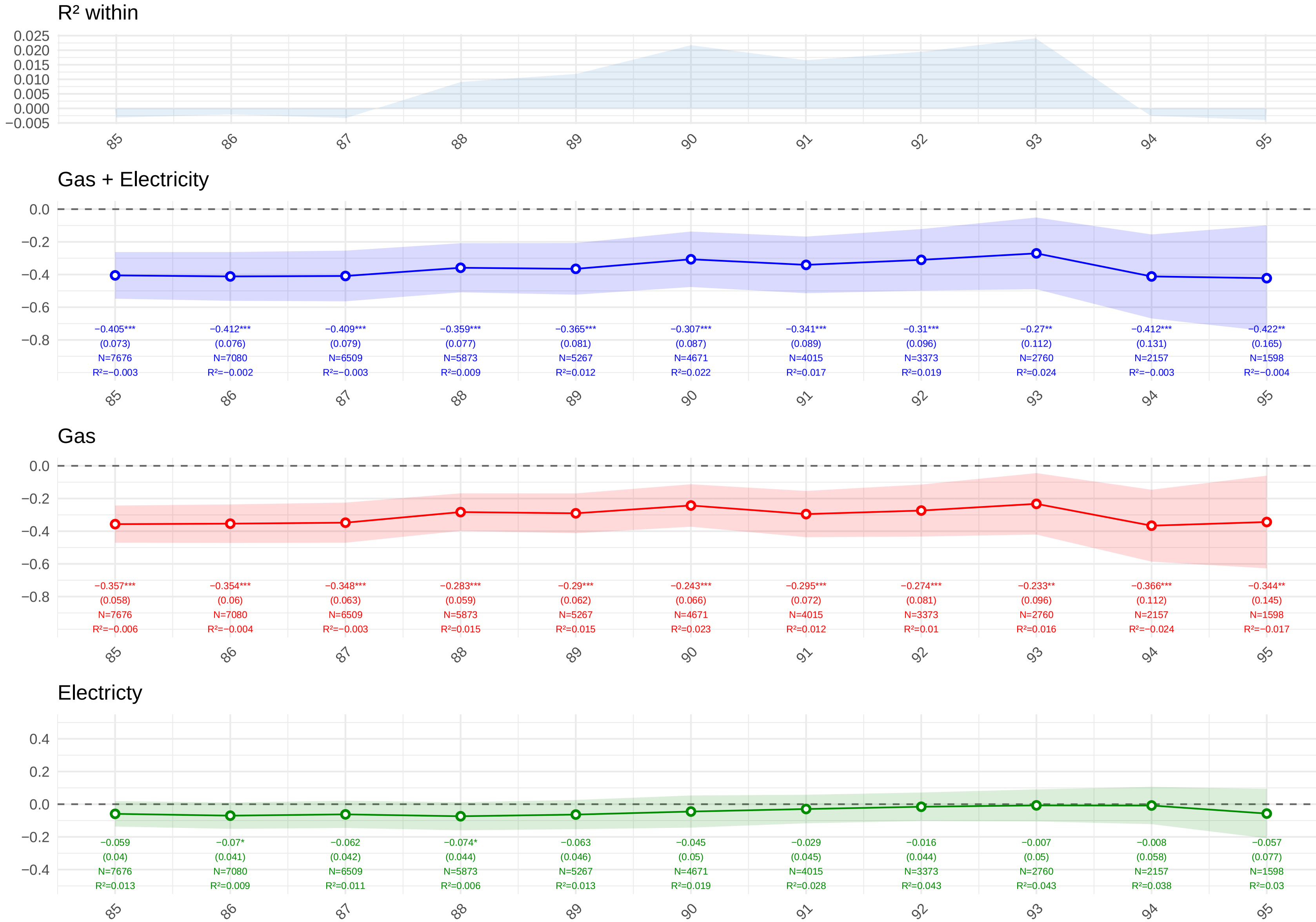


Figure B1: **IV estimates across exclusion thresholds (85–95%).** Each panel reports the 2SLS-FE coefficient on fitted expected savings (in kWh) for varying exclusion thresholds, separately for total energy consumption (*Gas + Electricity*), natural gas alone (*Gas*), and electricity alone (*Electricity*). The top-left panel plots the within-R^2 associated with each electricity specification. Each subplot displays the point estimate with 95% confidence interval, clustered standard errors at the neighbourhood level in parentheses, the number of observations (N), and the within-R^2. The vertical dashed line at the 90% threshold corresponds to the baseline specification reported in Table 3 and 4. Within-R^2 values turn negative at the extremes of the threshold range (below 87% and above 93%), guiding the choice of the 90% baseline. Significance levels: $^{**}p < 0.01$, $^{***}p < 0.001$.

TWFE Weights Diagnosis. In a second robustness check, we assess whether the two-way fixed-effects estimator assigns problematic negative weights to some average treatment effects on the treated (ATTs), following Chaisemartin and d'Haultfœuille 2020. Their critique applies to the OLS moment conditions of the TWFE estimator; our preferred 2SLS-FE strategy addresses a distinct concern (endogeneity of the treatment variable) and is not the target of this test. We therefore apply the diagnosis to the naive OLS-FE specification, which nonetheless serves as an important benchmark reported alongside the IV estimates in Table 3.

The test is particularly relevant in our setting for two reasons. First, the treatment variable defined as cumulative neighbourhood-level installations is continuously distributed and monotonically increasing over time: every neighbourhood accumulates some degree of treatment in every period, leaving no *never-treated* or *not-yet-treated* group to serve as a clean comparison. In this continuous, ever-treated design, TWFE is forced to construct implicit comparisons between more- and less-treated units, potentially assigning negative weights to (neighbourhood, year) cells whose ATT enters the aggregate coefficient with a sign opposite to its true effect. Second, even though the IV strategy is our preferred specification, the large gap between the OLS ($\hat{\beta} = -0.011$) and 2SLS ($\hat{\beta} = -0.307$) coefficients could in principle reflect either attenuation from classical measurement error, i.e., our favored interpretation, or TWFE aggregation artefacts inflating the negative weights on some ATTs. The CDH2020 diagnosis allows us to evaluate and rule out the latter.

Table B1 reports the results. Out of 13,487 ATTs that receive non-zero weight in the TWFE regression, 5,099 receive a positive weight (summing to +1.089) and 8,388 receive a negative weight (summing to -0.089). The share of total absolute weight attributable to negative weights is thus approximately 8.2%, and the minimum value of the sensitivity parameter $s(\Delta)$ compatible with a treatment effect of the opposite sign in all (neighbourhood, year) cells is 0.046—four times the magnitude of the treatment effect. The negative weights are therefore reassuringly small in magnitude and unlikely to account for the attenuation of the OLS estimate relative to the IV. This supports our interpretation that the OLS–IV gap reflects endogeneity and measurement error rather than TWFE aggregation bias.

	ATTs	S weights
Positive weights	5,099	1.0893
Negative weights	8,388	−0.0893
Total	13,487	1.0000

Notes. TWFE coefficient $\hat{\beta}_{\text{OLS}} = -0.0114$. Min. $s(\Delta)$ compatible with $\hat{\beta}_{\text{OLS}}$ and $\Delta_{\text{TR}} = 0$: 0.0058. Min. $s(\Delta)$ compatible with a treatment effect of opposite sign in all (g,t) cells: 0.0462. 13,491 (g,t) cells receive the treatment; 4 cells receive a weight of zero. Source: `TwoWayFEWeights` R package Chaisemartin and d'Haultfœuille 2020.

Table B1: **TWFE weight decomposition Chaisemartin and d'Haultfœuille 2020.** The table reports the number of ATTs and the sum of standardised weights (S weights) assigned by the naive OLS-FE estimator to (neighbourhood×year) cells with positive and negative weights.

C Accounting for Cross-Subsidies

Overview. A share of CEE-supported retrofit projects also received a fiscal transfer through either the tax credit for energy transition (CITE) or MaPrimeRénov' (MPR). Because this transfer, funded by taxpayers, enters the household cost function $c(x)$, it must be netted out to recover the net cost of abatement. This appendix details the construction of the overlap share π and the average transfer s per MWh of projected energy savings.

Overlap share π. We draw on the SDES report on subsidised residential energy renovations (ONRE 2022), which reports projected annual energy savings annually funded over the 2016–2021 period by the CEE program and the two public mechanisms (MPR and CITE). The report also identifies savings generated by investments that received support from both the CEE program and CITE/MPR simultaneously. Combining this information with data available in ONRE (2022) which gives the decomposition into mutually exclusive categories, we are able to compute the

$$\pi = \frac{E^{CEE+S}}{E^{CEE}}$$

where E^{CEE} denotes total CEE savings including doubly-supported projects. is not published directly by SDES using inclusion-exclusion. Over 2018–2021, doubly-supported savings amount to 5,112,747 MWh out of 24,648,251 MWh of total CEE savings, yielding $\hat{\pi} = 20.74\%$.

Average transfer s. I compute the average fiscal expenditure per MWh of energy savings generated by CITE- and MPR-supported projects. Annual energy savings under CITE and MPR are drawn from (ONRE 2022). CITE expenditure CITE_t comes from the Public Finances General Directorate data (Direction générale des Finances publiques 2022); MPR expenditure MPR_t comes from (Service des données et études statistiques 2024). Total fiscal expenditure in year t is $C_t = C_t^{\text{CITE}} + C_t^{\text{MPR}}$.

To express energy savings in lifelong savings, i.e., the unit in which CEE obligations are priced, we apply the standard CEE discounting formula at a 4% rate:

$$\tilde{E}_t^{\text{CITE+MPR}} = \left(E_t^{\text{CITE}} + E_t^{\text{MPR}}\right) \times \frac{1-(1.04)^{-L_t}}{1-(1.04)^{-1}}$$

where L_t is the conventional investment lifetime used in the CEE scheme. I take L_t as the expenditure-weighted average lifetime of CEE-certified works in year t, computed from the CEE microdata accessed through the CASD, the same source used in the econometric analysis. This ensures consistency with the pricing metric of the scheme, and embeds the maintained assumption that the average investment lifetime of works in the CITE/MPR–CEE intersection equals that of all CEE-supported works in the same year.

The average transfer is then computed over the pooled 2018–2021 period:

$$s = \frac{\sum_t C_t}{\sum_t \tilde{E}_t^{\text{CITE+MPR}}} = \frac{6{,}439{,}558{,}939}{171{,}885{,}146} \approx €\,37.17 \text{ per MWh cumac}$$

Interpretation. As discussed in the main text, πs measures the average fiscal transfer per MWh of CEE-certified savings. Converted into CO_2e units via the emission factors and coefficient estimates described in the main text, this correction accounts for approximately half the headline average abatement cost estimate, reflecting the substantial role of the cross-subsidy over the period.

D Calibration of program-induced energy price increases

By definition, the price increase rate of fuel f in year t, denoted by $\delta_{f,t}$, satisfies

$$m_{f,t} = (1+\delta_{f,t})\, m_{f,t}^0. \tag{13}$$

The value of $\delta_{f,t}$ is observed for the year 2019. Our objective is to recover the corresponding price increase rates for electricity and natural gas in 2018 and 2020.

To do so, we express the fuel price as a function of the certificate price:

$$m_{f,t} = m^0_{f,t} + \lambda_f p_t,$$

where λ_f denotes the fuel-specific pass-through rate, assumed to be constant over the study period. Combining this expression with Eq. (13) and rearranging yields

$$\lambda_f = \frac{m_{f,t}}{p_t} \left(\frac{\delta_{f,t}}{1 + \delta_{f,t}} \right). \tag{14}$$

We compute λ_f using observed certificate prices and fuel prices in 2019, together with program-induced price increases of $\delta_{g,2019} = 0.461$ for natural gas and $\delta_{e,2019} = 0.316$ for electricity.

Finally, we invert Eq. (14) to recover the price increase rate:

$$\delta_{f,t} = \frac{\lambda_f \, p_t}{m_{f,t} - p_t}.$$

We apply this expression to the years 2018, 2020 and 2021 to obtain the corresponding values of $\delta_{f,t}$. The resulting estimates are reported in Table C1.

Table C1: Estimated parameters

f	e	g
λ_f	0.7106	0.4100
$\delta_{f,\,2018}$	0.0250	0.0352
$\delta_{f,\,2020}$	0.0337	0.0621
$\delta_{f,\,2021}$	0.0347	0.0466